Spectral–Spatial–Frequency Geometry Adaptive Network for Hyperspectral Image Classification

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***Abstract*—The key to hyperspectral image (HSI) classification lies in the effective integration of spatial and spectral features. Furthermore, existing deep learning methods primarily operate in the spatial and spectral domains, often overlooking the potential of the frequency domain to capture global structural patterns and edge details. Because of this, a novel spectral–spatial–frequency geometric adaptive network (SFGANet) is proposed here. Specifically, a dual-domain feature modulator (DDFM) is introduced, which enhances the representation of details in the spatial domain through spatial–channel joint calibration while dynamically modulating components in the frequency domain to suppress noise and enhance task-relevant global geometric structures and edge textures, ultimately achieving the complementary integration of fine spatial features and global frequency features. Concurrently, a spectral band gating module (SBGM) is designed to enhance discriminative spectral features. To achieve deep synergy and complementarity of spatial and spectral branching features, an adaptive geometric interaction block (AGIB) is developed, which consists of geometry semantic synergistic attention (GSSA) and a feedforward network (FFN) with geometric enhancement capabilities. This architecture achieves a learnable, adaptive balance between cross-branch semantic interaction and local structure refinement. Furthermore, it employs a tokenization strategy to compress the high-dimensional features of the two branches into compact semantic tokens for classification prediction. In addition, the architecture effectively fuses the classification predictions from both branches using dynamic decision fusion (DDF), thereby mitigating the issue of overfitting with few labeled samples. Experiments were conducted on four datasets, and the results demonstrate that the proposed SFGANet significantly outperforms existing classification methods, thereby proving the model's effectiveness and superiority in HSI classification with limited training data.**

***Index Terms*—Adaptive geometric interaction block (AGIB), decision fusion, dual-domain feature modulator (DDFM), few labeled samples, hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL imaging technology enables the precise inversion of the physical and chemical properties of surface materials through fine-grained sampling across continuous narrow bands of the electromagnetic spectrum and has become an indispensable core technology in modern Earth observation [1]. Unlike traditional multispectral imagery, hyperspectral images (HSIs) are represented as 3-D data cubes that not only contain rich spatial context but also provide a continuous spectral response curve for each pixel, comprising dozens or even hundreds of spectral bands [2]. This highly distinctive "spectral fingerprint" makes it possible to investigate subtle material differences for surface cover types [3]. Because of its inherent superior spatial–spectral synergy, hyperspectral imaging technology exhibits significant potential in numerous remote sensing tasks, such as object detection [4], [5], spectral unmixing [6], [7], change detection [8], [9], and image enhancement [10], [11]. In particular, HSI classification [12], [13], as a cornerstone for understanding and interpreting remote sensing scenes, directly underpins a wide range of critical real-world applications [14], including precision agriculture [15], urban planning [16], environmental assessment [17], and disaster monitoring [18], by assigning precise semantic labels to individual pixels. However, the limited availability of labeled training samples poses a major challenge to discriminating between different land cover types from such high-dimensional hyperspectral data. To address this issue, it is imperative to develop robust and effective techniques that leverage available samples and extract discriminative features to achieve accurate land cover classification. Accordingly, this problem has continuously garnered extensive attention and in-depth research from the academic community.

In the early phase of HSI classification research, the majority of mainstream classification algorithms centered on exploiting the spectral characteristics of HSI data. Classical methods used for analysis included support vector machines (SVMs) [19], random forests (RFs) [20], principal component analysis (PCA) [21], and logistic regression [22]. Nevertheless, the application of these conventional approaches generally resulted in suboptimal classification accuracies as they inherently lack the capacity to capture fine-grained details within the imagery. In addition, these methods are plagued by further inherent limitations: heavy reliance on prior knowledge, suboptimal feature representation, and weak generalization capability, which collectively make it challenging to achieve a desirable tradeoff between feature discriminability and model robustness [23].

In recent years, the explosive advancement of deep learning techniques has attracted tremendous attention within the remote sensing and image processing communities. Leveraging its superior capacity for automating hierarchical feature learning and extraction, deep learning rapidly emerged as the primary driver advancing HSI classification. In contrast

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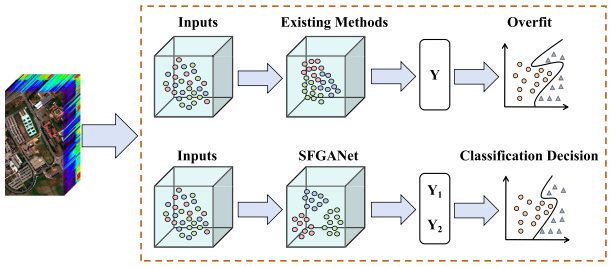


Fig. 1. Existing methods are inherently prone to overfitting in few-shot scenarios, while the proposed method can conduct in-depth mining of spatial–spectral–frequency features and mitigate the overfitting tendency through decision fusion.

to conventional machine learning approaches, deep learning effectively overcomes the inherent bottleneck posed by the limited representational capacity of handcrafted features, yielding a marked improvement in classification performance. Currently, the dominant deep learning paradigms for HSI classification fall into two well-established branches. The first is methods based on convolutional neural networks (CNNs) [24], [25], which excel at exploiting local spatial patterns and, thus, effectively capture the fine textural and geometric details of ground objects. The second is the emerging Transformer-centric architectures [26], [27], which leverage the self-attention mechanism to break through the inherent limitation of fixed receptive fields in CNNs and exhibit superior capability for global context modeling. These two branches of approaches exhibit inherently complementary strengths, which collectively drove the paradigm shift of HSI classification from shallow feature matching to deep semantic understanding.

Beyond the two dominant deep learning paradigms outlined above, a range of other deep learning-driven methods have also been explored for HSI classification, including approaches based on diffusion models [28], Mamba architectures [29], [30], and other emerging deep learning frameworks [31], [32], [33]. HSIs are inherently characterized by continuous spectral bands and intricate spatial structures, which pose inherent challenges for high-dimensional feature modeling in HSI classification tasks. On the one hand, diffusion models, leveraging their inherent iterative stepwise denoising mechanism, can effectively capture and fit the complex joint spatial–spectral distribution of such high-dimensional HSI data [34]. On the other hand, the state-of-the-art emerging Mamba architecture circumvents the computational bottleneck intrinsic to conventional self-attention mechanisms, enabling efficient modeling of global dependencies in ultralong sequences with linear computational complexity. This allows the architecture to substantially boost the representation capacity of global contextual features without incurring a significant increase in computational overhead [35]. However, in practical remote sensing acquisition scenarios, obtaining high-fidelity pixel-level ground truth labels for hyperspectral imagery incurs prohibitive labor and time costs, and further requires extensive expert domain knowledge, as well as field survey and validation [36]. As shown in Fig. 1, bridging the gap between sophisticated deep models and scarce labeled samples, and developing advanced algorithms capable of robustly extracting discriminative features while maintaining strong classification performance under few-shot constraints, emerged as the core bottleneck restricting the real-world deployment of HSI classification techniques [37].

To this end, a novel spectral–spatial–frequency geometric adaptive network (SFGANet) is proposed here. While frequency-domain methods have been employed in HSI classification to capture global structural patterns and long-range dependencies, most existing works either process frequency features independently or merge them with spatial–spectral features via concatenation. Consequently, they lack a deep adaptive cross-domain synergy mechanism and fail to effectively preserve local geometric details [38], [39], [40]. To address these issues, SFGANet innovatively constructs a unified spectral–spatial–frequency feature optimization framework, achieving the end-to-end synergistic enhancement of multidomain features through tailored adaptive modules. At the front end of the network, the model incorporates a spectral band gating module (SBGM), which adaptively filters high-value information from the continuous spectral range through dynamic weight allocation, focusing on the most discriminative combinations of spectral bands. Building on this, the model introduces a dual-domain feature modulator (DDFM) to achieve deep collaborative processing between the spatial and frequency domains, enabling the model to dynamically customize feature extraction strategies based on the textural characteristics of different input samples. To further enhance the deep alignment of cross-modal features, SFGANet embeds an adaptive geometric interaction block (AGIB) within its dual-branch architecture. Through efficient information exchange across branches, this module not only enhances the consistency of global semantics in complex nonlinear manifold spaces but also preserves and reinforces local geometric structures. Finally, during a high-level semantic decision-making stage, the model employs a learnable tokenization strategy to map the fused high-dimensional features into a compact semantic subspace. By optimizing the global contextual relationships among tokens, it generates highly refined and highly discriminative feature vectors. Furthermore, DDF generates adaptive fusion weights for the spatial and spectral branches of each pixel, thereby performing weighted fusion of the classification results from both branches. In summary, SFGANet achieves deep closed-loop fusion of global high-dimensional semantics and local microscopic geometric topology while suppressing overfitting in small-sample scenarios, thereby significantly improving the clarity of classification decision boundaries and overall performance in complex hyperspectral scenes, as shown in Fig. 1. The main contributions of this work are summarized as follows.

1) A novel SFGANet is proposed. To address the inherent limitation that most existing methods are confined to 2-D spatial–spectral representations, this article constructs a unified feature interpretation framework that deeply integrates spectral, spatial, and frequency dimensions. While effectively mitigating the overfitting of deep learning models in few-shot scenarios, the proposed network significantly improves the discriminative capability of the model for complex hyperspectral scenes.

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2) A novel DDFM module is proposed, which constructs an adaptive feature fusion paradigm that synergizes spatial local refinement and frequency-domain global modulation. By dynamically balancing the contributions of frequency-domain global context, spatial local features, and channel attention, it overcomes the inherent limitation of traditional frequency-domain methods, which are prone to introducing artifacts or losing local details. The module adaptively suppresses high-frequency noise while enhancing discriminative textural features, thus achieving efficient synergistic representation of spatial structures and frequency spectrum information in HSIs.

3) A novel AGIB is proposed, which achieves cross-branch spatial–spectral feature interaction through geometry semantic synergistic attention (GSSA) that simultaneously models content semantic context and spatial geometric context, while a depthwise convolution feedforward network (FFN) focuses on enhancing local geometric details. Furthermore, a learnable scalar parameter is introduced to enable the model to reach a dynamic optimal balance between cross-branch semantic sharing and geometric detail refinement.

4) A novel DDF strategy is proposed. It fully preserves the independent discriminative advantages of the dual-branch architecture while maximally exploiting the classification potential of joint spatial–spectral information. By adopting independent classifiers for the two branches and an end-to-end learnable dynamic weight generation mechanism, it achieves adaptive and optimal fusion of spatial and spectral classification results, effectively avoiding semantic conflicts and information redundancy caused by early feature fusion.

The rest of the article is organized as follows. Section II reviews related work. Section III describes the structure of the proposed model and each functional module in detail. Section IV describes the datasets used in the model, the relevant configurations for the experiments, and the experimental results. Section V gives conclusions on the overall work.

II. Related Work

In this section, we give a summary of the main approaches in the HSI classification work. This section focuses on the following three approaches: methods based entirely on CNNs for HSI classification, methods based entirely on Transformers for HSI classification, and methods based on a hybrid CNN-Transformer for HSI classification.

A. Methods Based on CNNs for HSI Classification

Given the high efficiency of CNNs in spatial feature extraction, along with their excellent properties of convolution operations, such as strong local receptive fields and nonlinear representation capability, CNN-based methods have been widely applied in the field of HSI classification. As one of the pioneering explorations introducing deep learning into this field, Hu et al. [41] proposed a classic 1D-CNN classification framework. This network elaborately constructs a five-layer cascaded architecture consisting of convolutional layers, pooling layers, and fully connected layers, which is designed to directly perform deep feature mapping on the 1-D spectral sequences of hyperspectral pixels. By treating each pixel as an independent individual, this processing mechanism breaks the continuity of ground object distribution and local spatial topological dependencies, presenting an obvious blind spot in spatial context information extraction. To break through the aforementioned bottleneck of spatial feature representation, 2D-CNNs, represented by the research of Zhao and Du [42], were formally introduced to compensate for the deficiencies of 1D-CNNs in spatial modeling for HSI classification. By adopting a spatial patch-based input strategy, 2D-CNN can utilize 2D-CNN kernels to effectively capture the local texture, along with the shape and geometric structure features of the central pixel and its adjacent neighborhoods, thus significantly improving the model's discriminative capability for complex ground object boundaries. To fully exploit the 3-D structural advantages of hyperspectral data cubes, He et al. [43] proposed a multiscale 3-D deep CNN (M3D-CNN) framework, which innovatively introduces multiscale 3-D convolutional kernels. This design enables the model to capture spatial–spectral joint features synchronously and dynamically directly from raw hyperspectral data patches without any dimensionality reduction preprocessing. As the network depth increases, the model usually faces inherent optimization bottlenecks of deep networks, such as vanishing gradients and performance degradation. To address this issue, Zhong et al. [44] constructed a spectral–spatial residual network (SSRN). This framework, in an innovative way, deeply couples the residual learning paradigm with 3-D convolution operations and effectively breaks the information transmission barrier in deep networks by introducing continuous skip connections, thereby extracting more discriminative deep spatial–spectral cascaded features with richer hierarchical information. Zhu et al. [45] further developed a residual spatial–spectral attention network (RSSAN). Instead of processing spatial and spectral features in isolation, this architecture, in an innovative fashion, deeply embeds the joint spatial–spectral attention module into residual learning blocks. This coupled optimization strategy not only inherits the stability of residual topology in deep information transmission but also realizes efficient focusing and enhanced extraction of critical discriminative features in HSIs through dynamic spatial–spectral weight assignment. However, CNN-based methods still have inherent limitations, such as restricted capacity in modeling global context information and difficulty in capturing long-range dependencies.

B. Methods Based on Transformer for HSI Classification

Building on the tremendous success of Transformer architectures in the field of natural language processing (NLP), Dosovitskiy et al. [46] pioneered the introduction of this architecture into the computer vision (CV) domain and proposed the Vision Transformer (ViT) model. By splitting an image into a series of nonoverlapping patches and linearly embedding them into sequential tokens, ViT captures long-range dependencies between patches via a self-attention mechanism. This innovation provides a novel modeling perspective for CV tasks and exhibits significant advantages, especially in global context information processing. The Transformer architecture effectively compensates for the shortcomings of

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CNNs in long-range dependency modeling. By dynamically aggregating spectral features from different spatial positions via the self-attention mechanism, it further enhances the model's discriminative capability for land cover categories in complex hyperspectral scenes. Researchers have subsequently introduced the Transformer architecture into HSI classification tasks. Hong et al. [47] were the first to formally bring the Transformer structure into the HSI classification field, proposing the spectralFormer model. This model breaks the dependence of traditional networks on local receptive fields and innovatively treats hyperspectral pixels and their adjacent bands as a continuous spectral sequence. By designing a groupwise spectral embedding mechanism, it realizes effective capture of the global sequence attributes of hyperspectral data under the Transformer framework. Furthermore, to mine deep correlations along the spectral dimension, Qing et al. [48] proposed an end-to-end transformer model entitled SATNet. By fusing a customized spectral attention mechanism with a multihead self-attention module equipped with trainable embedding vectors, SATNet can not only efficiently and synergistically extract highly discriminative spatial–spectral features but also further strengthen the network's ability to model long-range spectral dependencies in high-dimensional hyperspectral data. To address the core challenge that existing models struggle to balance spatial and spectral feature modeling, He et al. [49] proposed a dual-branch pure transformer model. The model constructs two independent branches, namely, Spatial Transformer (Spa-TR) and Spectral Transformer (Spe-TR). Specifically, for spatial feature extraction, Spa-TR innovatively introduces window partition and shifted window mechanism, which strictly confines self-attention computation within local windows and cross-window interactions, thus realizing simultaneous perception of local spatial context and global topological structure. Subsequently, Yu et al. [50] proposed a multilevel spatial–spectral transformer network (MSTNet). By incorporating learnable position encoding, MSTNet directly maps hyperspectral data into sequential tokens, which are then uniformly fed into the Transformer encoder for multiscale feature learning. Although these Transformer-based networks have distinct advantages in modeling long-range feature dependencies, their capability for spatial feature extraction remains limited, resulting in insufficient utilization of HSI data.

C. Methods Combining CNNs and Transformer Networks

To fully exploit the complementary advantages of CNNs and Transformers, hybrid CNN-Transformer paradigms have been developed accordingly. By combining the potent local feature extraction of CNNs and the long-range global modeling capacity of Transformers, such methods hold great potential for further elevating the classification performance. In early exploratory work, He et al. [51] constructed a spatial–spectral transformer (SST) model and deeply exploited the deep long-range dependencies between the continuous spectral bands of hyperspectral data by introducing a densely connected self-attention architecture. Sun et al. [52] proposed a spatial–spectral feature tokenization Transformer (SSFTT). The front end of their framework relies on cascaded 3-D and 2-D convolutions to capture shallow spatial priors with strong inductive bias, which are subsequently mapped into high-level semantic tokens and fed into the Transformer encoder. The design successfully realizes the complementary advantages of CNNs in local detail extraction and Transformers in global receptive field modeling. Following the research line of enhancing spatial geometric representation, Roy et al. [53] further proposed a spatial–spectral morphological attention Transformer (MorphFormer). The study embeds the dilation and erosion operations from mathematical morphology into the self-attention module, enabling it to more accurately characterize the local topological structures of pixels and their spatial morphological correlations in complex hyperspectral scenes. Furthermore, addressing the interband correlations inherent in hyperspectral sequences, Qi et al. [54] proposed a pioneering global–local 3-D convolutional Transformer network. By meticulously analyzing the complex local spectral topological structures within individual bands and collaboratively modeling global sequence features across bands, this approach achieves the in-depth decomposition and fusion of macro- and micro-spatial–spectral information. To alleviate the potential performance degradation problem in deep networks, Sun et al. [55] proposed a memory-augmented SST (MASSFormer), which leverages a memory token generator to convert the extracted features into memory tokens for prior knowledge storage and realizes in-depth interaction of spatial–spectral information through a memory-augmented encoder. Zhou et al. [56] proposed a dual-branch convolutional Transformer network (DCTN), which, respectively, adopts CNN and Transformer to extract spatial and spectral features in parallel and, in an innovative way, embeds an efficient interactive self-attention mechanism in the Transformer branch. In addition, Dun et al. [57] proposed the dual-branch token selection fusion network (DBTSFN). Through a collaborative architecture combining a spatial–spectral cascaded strategy with a gated-enhanced token selection module, it achieves refined collaborative modeling of complementary spatial and spectral information. Furthermore, by employing a dual-branch adaptive classification decision fusion mechanism, it mitigates model overfitting in scenarios with limited labeled samples.

In addition to the aforementioned CNN and Transformer architectures, graph neural networks (GNNs) have recently demonstrated exceptional potential in HSI classification due to their inherent flexibility in modeling non-Euclidean spatial structures and complex land cover topological dependencies. For instance, to efficiently capture rich spatial–spectral features, multiscale receptive fields have been innovatively integrated into graph attention neural networks [58]. Similarly, adaptive path aggregation graph neural frameworks have been proposed to leverage multireceptive fields for enhanced feature representation and structural preservation [59]. Furthermore, exploring the paradigm of unsupervised learning, a lightweight self-supervised low-pass contrastive graph clustering (SLCGC) network has been developed to mine intrinsic feature distributions without relying on extensive labeled data [60]. To mitigate noise interference and effectively aggregate local neighborhood information, graph convolution mechanisms equipped with adaptive filters

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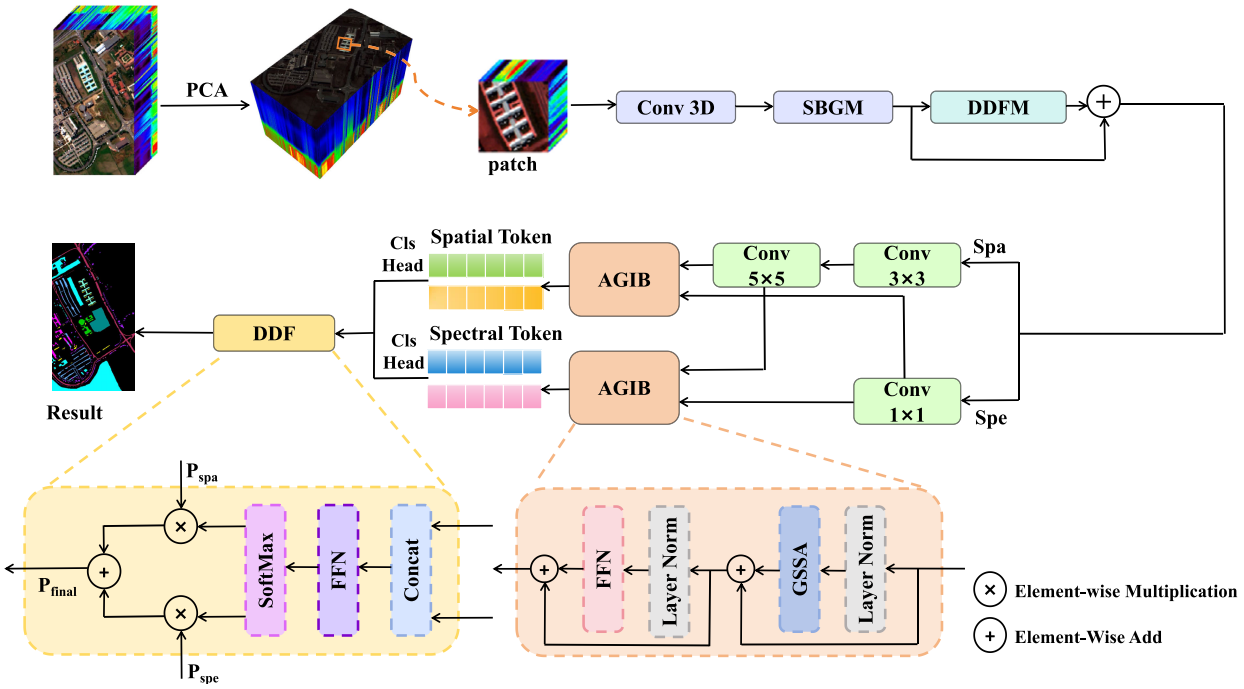


Fig. 2. Overall architecture of the proposed SFGANet for HSI classification.

and aggregator fusion (AF2GNN) have also been explored [61]. While these GNN-based methods excel in topological modeling, extracting discriminative spatial–spectral features in low-data-volume scenarios remains a critical challenge, which inspired the cross-domain synergistic design of our proposed SFGANet.

Although the aforementioned works have demonstrated some progress, three major bottlenecks remain, limiting further improvements in model performance: First, there is insufficient efficient synergy between spatial and spectral features, making it difficult to mitigate overfitting with limited samples and failing to fully integrate the local context extraction capabilities of CNNs with the long-range dependency modeling advantages of Transformers. Second, existing methods are mostly confined to the spatial or spectral domains, severely neglecting the global complementary prior information inherent in the frequency domain. Third, existing spatial–spectral interaction mechanisms lack adaptive protection for local geometric structures. Traditional hybrid networks rely on simple parallel branches and feature concatenation, making it difficult to achieve deep alignment of cross-domain features in complex nonlinear spatial–spectral manifold spaces.

III. Proposed Method

In this section, we will describe the overall architecture of the SFGANet method, including detailed information on the SBGM, DDFM, AGIB, and DDF modules.

A. Overview of SFGANet

A schematic of the proposed SFGANet architecture is shown in Fig. 2. The SFGANet consists of four main components: SBGM, DDFM, AGIB, and DDF. First, SBGM adaptively recalibrates the importance of spectral bands

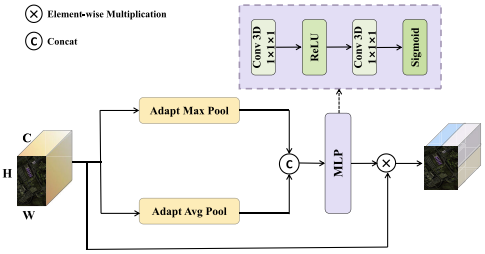


Fig. 3. Illustration of the SGBM module.

through 3-D global pooling and an MLP-based gating mechanism, thereby effectively enhancing the network's ability to distinguish spectral features. The DDFM combines a Fourier transform branch for global frequency-domain modeling with a partial channel attention mechanism for local spatial refinement, enabling the network to capture both long-range dependencies and fine-grained textures simultaneously. AGIB consists of a GSSA and an FFN with local spatial awareness. The GSSA facilitates cross-modal interaction between the spatial and spectral branches through a cross-attention mechanism with dynamic position encoding. After the GSSA completes coarse-grained cross-modal alignment, the FFN performs local refinement and enhancement of features to mitigate feature loss. Finally, by employing the DDF strategy, unique spatial–spectral fusion weights are dynamically generated for each pixel based on input features. These weights are then used to adaptively fuse the classification decisions from the spectral and spatial branches, thereby effectively mitigating the risk of overfitting in low-data-volume scenarios.

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B. Spectral Band Gating Module

Prior to the introduction of the SBGM module, 3-D convolution is capable of processing both spatial and spectral dimensions simultaneously, effectively capturing spatial–spectral joint features in HSIs. However, it has a key limitation in that it applies the same convolution kernel to all spectral bands, failing to distinguish the relative importance of different bands. Therefore, a lightweight SBGM is proposed here. As shown in Fig. 3, SBGM uses a self-learning mechanism to adaptively assign weights to each spectral band. First, adaptive average pooling and adaptive max pooling are employed to comprehensively capture spectral statistical information. Subsequently, a lightweight MLP generates channel weights. Applying these adaptive weights to the input features effectively highlights discriminative bands while suppressing irrelevant ones, thereby mitigating intraclass spectral variability, enhancing interclass distinguishability, and providing more discriminative representations for subsequent feature extraction.

Let $X \in \mathbb{R}^{C\times D\times H\times W}$ denote the input 3-D hyperspectral feature tensor, where C is the number of input channels, D is the number of spectral bands, and H and W are the spatial resolutions. First, to extract global feature descriptions for each spectral band, we apply global average pooling (GAP) and global maximum pooling (GMP) in parallel. GAP aggregates global spatial distribution trends across the spectral band, serving to smooth out noise, while GMP extracts the most significant response features within the band, preserving key spectral details essential for object recognition. Using them in parallel allows us to fully leverage the strengths of both methods, enabling feature extraction for each spectral plane from different perspectives

$$Y_{\text{avg}} = \text{AvgPool}_{3\text{D}}(X) \tag{1}$$

$$Y_{\text{max}} = \text{MaxPool}_{3\text{D}}(X). \tag{2}$$

Next, we concatenate the outputs of GAP and GMP to obtain the fused feature $Y_{\text{cat}} \in \mathbb{R}^{2\times D\times 1\times 1}$. This concatenated feature vector combines two key pieces of information: global spatial distribution trends and key spectral details. Subsequently, we use a sequence of 3-D convolutional layers with a bottleneck structure to capture the nonlinear dependencies between bands

$$Z = \sigma\left(\text{Conv}_{3\text{D}}^{1\times 1\times 1}\left(\text{ReLU}\left(\text{Conv}_{3\text{D}}^{1\times 1\times 1}(Y_{\text{cat}})\right)\right)\right). \tag{3}$$

In particular, the ReLU layer introduces nonlinearity to assist the network in learning complex spectral interactions, while σ represents the sigmoid activation function, which normalizes the output to the band weight coefficients $\mathbf{Z} \in \mathbb{R}^{1\times D\times 1\times 1}$. Finally, the generated weight vector Z is applied to the original input features X. Through elementwise multiplication, this achieves dynamic gain adjustment across spectral dimensions, thereby enhancing the discriminative power of the spectral representation during the feature extraction stage

$$X_{\text{out}} = Z \otimes X. \tag{4}$$

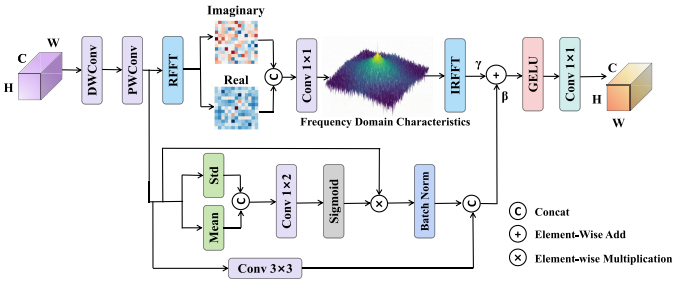


Fig. 4. Illustration of the DDFM module.

C. Dual-Domain Feature Modulator

Once SBGM completes the spectral calibration, we flatten the 3-D spectral features into a 2-D feature map. At this stage, the feature map contains both the core spectral semantics derived from SBGM calibration and the original spatial texture and structural information. Now, the SBGM is confronted with two emerging intractable challenges. First, in flattened high-dimensional features, critical local information for classification is diluted, including the edges, textures, and small target contours of ground objects, and these features are precisely the core basis for distinguishing different objects with the same spectrum. Second, conventional convolution can only capture local neighborhood information and cannot efficiently model the global spatial distribution patterns of large-range homogeneous ground objects. Therefore, to simultaneously account for both local detail perception and the capture of global contextual information during feature extraction, DDFM is proposed, as shown in Fig. 4. This module employs a parallel dual-branch structure to modulate and enhance features in both the spatial and frequency domains. DDFM builds upon the spectral calibration performed by SBGM, serving as the central hub that bridges the two backend spatial–spectral branches. Together with SBGM, it forms a complete spatial–spectral feature optimization chain, ranging from spectral-domain calibration to spatial–spectral dual-domain enhancement. Its core design philosophy employs a dual-path parallel architecture combining a spatial local branch and a frequency-domain global branch, and then achieves adaptive aggregation of dual-domain features through learnable gated weights. The final output consists of high-quality spatial–spectral fusion features that combine strong spectral semantics, fine spatial details, and long-range dependencies across the entire domain.

Let $X_{\text{in}} \in \mathbb{R}^{C_{\text{in}}\times H\times W}$ denote the input features after dimensionality reshaping. To expand the feature capacity while extracting local spatial information, $\text{DWConv}_{3\times 3}$ is employed as a spatial feature extractor, which not only captures spatial neighborhood relationships within the local receptive field but also expanded the feature channel dimension from C_{in} to C_{h}. The subsequent pointwise convolution $\text{PWConv}_{1\times 1}$ does not alter the feature dimension. Instead, it is specifically responsible for cross-channel information interaction and fusion within the expanded high-dimensional space, providing a feature foundation rich in local semantics for subsequent

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frequency-domain modulation

$$X_{\text{local}} = \text{PWConv}_{1\times1}\left(\text{DWConv}_{3\times3}\left(X\right)\right). \tag{5}$$

After obtaining the refined feature X_{local}, the network feeds it into the frequency-domain global modulation branch and the spatial channel calibration branch for parallel processing. It first goes through the frequency-domain global modulation branch. Different ground object categories often face the core challenges of the same-spectrum different-object and same-object different-spectrum phenomena due to the complexity of spectral and spatial features. This dilemma can be effectively addressed via frequency-domain analysis: The low-frequency components accurately capture the overall contours and spatial position information of ground objects. On the other hand, the mid-frequency components encapsulate abundant structural texture features to discriminate fine details of ground object boundaries, while the high-frequency components focus on edge details and weak feature extraction, and simultaneously suppress noise interference. This multiband feature collaborative mechanism not only significantly enhances the model's discriminative capability for spectrally similar scenarios but also provides more robust feature representations for HSI classification. To capture global periodic patterns and long-range dependencies, we introduce the 2-D real fast Fourier transform (2D-RFFT) to map the spatial-domain feature X_{local} into the frequency domain

$$X_{\text{freq}} = \text{RFFT}\left(X_{\text{local}}\right). \tag{6}$$

After separating the signal into a real part $\mathbf{R} = \text{Re}(X_{\text{freq}})$ and an imaginary part $\mathbf{I} = \text{Im}(X_{\text{freq}})$, and concatenating them along the channel dimension, we apply a 1×1 convolution to modulate the signal. This allows the model to adaptively learn weights, enhancing low-frequency global components that are beneficial for classification while suppressing undesirable high-frequency components, thereby achieving cross-channel interaction between global frequency amplitude and phase—something that spatial-domain convolution cannot achieve

$$\left[\hat{\mathbf{R}}, \hat{\mathbf{I}}\right] = \text{Conv}_{1\times1}\left(\text{Concat}\left[\mathbf{R}, \mathbf{I}\right]\right). \tag{7}$$

Subsequently, the modulated complex representation is mapped back to the spatial domain via a 2-D real inverse Fourier transform (2D-IRFFT), thereby leveraging the global properties of the frequency domain to overcome the limitations of the convolution operation's receptive field

$$X_{\text{F}} = \text{IRFFT}\left(\hat{\mathbf{R}} + \text{j}\hat{\mathbf{I}}\right). \tag{8}$$

Next is the spatial channel calibration branch. To reduce computational redundancy, X_{local} is split into two parts based on the channel ratio: $X_{\text{p1}} \in \mathbb{R}^{C_1\times H\times W}$ and $X_{\text{p2}} \in \mathbb{R}^{C_2\times H\times W}$. One part leverages $\text{Conv}_{3\times3}$ to further enhance spatial details and capture local spatial structures. The other part adopts a statistically based channel attention module to perform adaptive interchannel feature calibration. This design simultaneously takes into account local spatial detail enhancement and global channel semantic calibration, and avoids the dilution of the spectral information calibrated by SBGM

$$X'_{\text{p1}} = \text{Conv}_{3\times3}\left(X_{\text{p1}}\right) \tag{9}$$

$$X'_{\text{p2}} = \text{BN}\left(A_{\text{c}} \otimes X_{\text{p2}}\right) \tag{10}$$

$$A_{\text{c}} = \sigma\left(\text{Conv}_{1\times2}\left(\text{Concat}\left[\text{Mean}\left(X_{\text{p2}}\right), \text{Std}\left(X_{\text{p2}}\right)\right]\right)\right). \tag{11}$$

In this approach, A_{c} represents the channel attention weights that combine mean and standard deviation. Some channels carry the spectral semantic information of key features, while others carry spatial texture information. This split design allows the simultaneous enhancement of both types of features at minimal cost, thereby strengthening the module's multiscale fusion capabilities. Together with SBGM, the design forms a two-tier calibration system that combines global spectral calibration with channel-level fine-tuning. Finally, the two components are recombined

$$X_{\text{S}} = \text{Concat}\left(X'_{\text{p1}}, X'_{\text{p2}}\right) \tag{12}$$

where X'_{p1} enhances the representation of local spatial features, while X'_{p2} enriches the distribution characteristics of channel weights using second-order statistics; the two complement each other to strengthen the discriminative channels. Finally, the frequency-domain output X_{F} and the spatial-domain output X_{S} are adaptively weighted and fused using the learnable scalar parameters γ and β and then restored to the original dimension via an activation function and linear projection

$$X_{\text{fuse}} = \gamma X_{\text{F}} + \beta X_{\text{S}} \tag{13}$$

$$X_{\text{DDFM}} = \text{Conv}_{1\times1}\left(\text{GELU}\left(X_{\text{fuse}}\right)\right). \tag{14}$$

The control parameters γ and β enable the network to dynamically balance global frequency context with local spatial details. Since different object types have varying requirements for global versus local information, these learnable control parameters allow the model to adaptively determine the optimal fusion ratio during training, thereby achieving a dynamic balance between global structure and local details. Finally, a residual connection is established with the output X_{out} of the SBGM to ensure that the original spectral features produced by the SBGM are not diluted by the transformations performed by the DDFM. This achieves complementarity between the original spectral features and the dual-domain enhanced features, thereby fundamentally preventing the loss of spectral semantic information.

D. Adaptive Geometry Interaction Block

As shown in Fig. 2, the hyperspectral fusion features obtained after spatial–frequency domain collaborative enhancement via DDFM are decoupled into spatial and spectral feature branches through two sets of distinct convolution operations: The spatial branch employs convolutions with varying receptive fields to capture the spatial geometric structure and global contextual information of ground objects, while the spectral branch uses 1×1 pointwise convolutions to focus on interband dependencies and discriminative spectral features. To achieve the deep collaboration, bidirectional complementation, and semantic alignment of heterogeneous spatial and spectral features, we propose the AGIB module. Centered on the GSSA mechanism as shown in Fig. 5, this module

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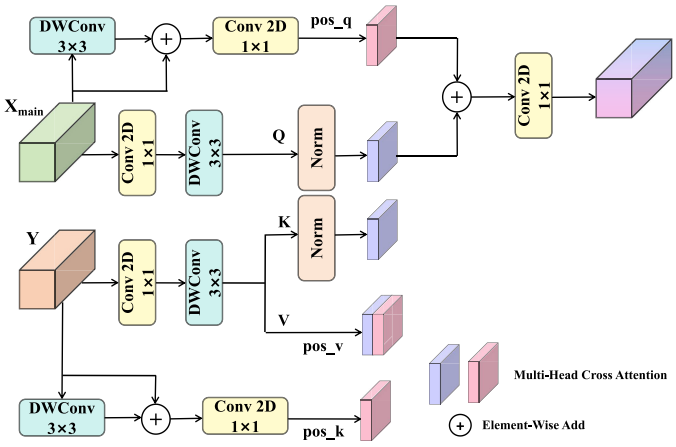


Fig. 5. Illustration of the GSSA module.

incorporates a dynamic position-aware scheme to ensure the spatial alignment of cross-modal features. Furthermore, it is equipped with an FFN that possesses both spatial perception capabilities. In addition, a set of learnable adaptive weights was introduced to balance the contributions of cross-modal interaction and local feature enhancement. Such a design guides the primary modality to accurately extract matched geometric structures and discriminative semantic information from the auxiliary modality, thereby fulfilling the bidirectional collaborative enhancement of spatial and spectral features.

Within the bidirectional interaction architecture of the overall network, we deploy two parallel AGIB modules: The first module uses spatial branch features as the primary modality and spectral branch features as the auxiliary modality, guiding spatial features to absorb highly discriminative spectral semantic information from spectral branches while suppressing interference from spatial background noise. The second module uses spectral branch features as the primary modality and spatial branch features as the auxiliary modality, introducing spatial geometric constraints to spectral features to mitigate classification confusion caused by same-object different-spectrum and same-spectrum different-object phenomena, ultimately providing more discriminative features for subsequent tokenization representation.

Let the primary modality input features be denoted as $X_{\text{main}} \in \mathbb{R}^{C\times H\times W}$ and the secondary modality input features as $Y \in \mathbb{R}^{C\times H\times W}$. In the network, $X_{\text{main}} \in \mathbb{R}^{C\times H\times W}$ and $Y \in \mathbb{R}^{C\times H\times W}$ are alternately fed as spatial and spectral branch features, respectively. To preserve local geometric structures in feature interactions, we use a cascaded architecture combining 1×1 convolutions and 3×3 DWConv to generate queries, keys, and values

$$Q = \text{DWConv}_{3\times 3}\left(\text{Conv}_{1\times 1}\left(X\right)\right) \tag{15}$$

$$K, V = \text{Chunk}\left(\text{DWConv}_{3\times 3}\left(\text{Conv}_{1\times 1}\left(Y\right)\right)\right). \tag{16}$$

In particular, the Chunk operation splits the features into two parts along the channel dimension. To stabilize the attention calculation, we apply L_2 normalization to the unfolded Q and K. The basic attention map based on content interaction is computed as follows:

$$A_{\text{base}} = \text{Softmax}\left(\tau \cdot \left(\hat{Q}\hat{K}^{\text{T}}\right)\right) V \tag{17}$$

where $\tau \in \mathbb{R}^{h\times 1\times 1}$ represents the learnable parameters in the multihead attention mechanism and h denotes the number of attention heads. In addition, a feature-driven dynamic spatial encoding mechanism is introduced, in which zero-padded DWConv is used to dynamically learn spatial representations directly from the current input features. This design not only accurately compensates for the spatial and geometric information lost in the attention mechanism, providing reliable spatial constraints for attention weight calculations, but also fully preserves the model's translation invariance. At the same time, it enables the collaborative modeling of spatial information and semantic content

$$P_X = X_{\text{main}} + \text{DWConv}_{3\times 3}\left(X\right) \tag{18}$$

$$P_Y = Y + \text{DWConv}_{3\times 3}\left(Y\right) \tag{19}$$

$$Q_{\text{pos}} = \text{Conv}_{1\times 1}\left(P_X\right) \tag{20}$$

$$K_{\text{pos}} = \text{Conv}_{1\times 1}\left(P_Y\right). \tag{21}$$

By combining the outputs of the semantic content cross-attention branch with those of the geometric position-aware attention branch, bidirectional collaborative modeling of semantic content associations and spatial geometric structures is achieved. Specifically, the semantic content attention branch focuses on matching high-level semantic similarities among features to uncover intrinsic content relationships across branches; the geometric position attention branch, based on input-driven dynamic position encoding, models the relative positional relationships and geometric structural constraints of pixels in the feature space, thereby accurately supplementing the spatial position information that is easily lost in global attention mechanisms. The elementwise addition of the two feature streams enables deep coupling of semantic information and geometric position information without compromising the intrinsic independent representation properties of each stream, endowing the fused features with both precise content discriminative capability and reliable spatial geometric perception capability. The fused features are transformed and calibrated through a channel-dimension linear projection layer. In this way, global integration and representation optimization of the channel dimension are completed, while the original spatial size is maintained, and GSSA features are finally output

$$A_{\text{pos}} = \text{Softmax}\left(Q_{\text{pos}}K_{\text{pos}}^{\text{T}}\right) V \tag{22}$$

$$X_{\text{GSSA}} = \text{Conv}_{1\times 1}\left(A_{\text{base}} + A_{\text{pos}}\right). \tag{23}$$

While traditional Transformer FFNs typically employ fully connected layers, an FFN capable of local spatial perception was designed. By utilizing DWConv, it models local spatial features while performing channel transformations, thereby achieving complementary feature representation between global and local scales in conjunction with global attention. Furthermore, to ensure the training stability of the deep network, normalization operations and a learnable fusion parameter α were adopted. This design adaptively balances the feature contributions of the global GSSA branch and the local perception FFN branch, dynamically regulates the feature

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flow between the global attention branch and the local branch, and, thus, realizes the adaptive collaborative fusion of the two branches

$$X' = X + \alpha \cdot \mathrm{GSSA}\left(\mathrm{LN}\left(X\right), \mathrm{LN}\left(Y\right)\right) \tag{24}$$

$$X_{\mathrm{out}} = X' + (1-\alpha) \cdot \mathrm{FFN}\left(\mathrm{LN}\left(X'\right)\right). \tag{25}$$

E. Dynamic Decision Fusion

After dual-branch independent feature extraction and cross-modal interaction, the network is required to convert the two streams of heterogeneous spatial–spectral information into a unified classification decision. To this end, a dynamic decision fusion (DDF) mechanism is designed at the output end of the network. Guided by the deep semantic features of the dual branches, this mechanism adaptively generates decision fusion weights for the spectral and spatial branches for each input sample, realizing pixel-level adaptive decision-level fusion. While retaining the independent discriminative advantages of the dual branches, it maximally exploits the classification potential of joint spatial–spectral information. As shown in Fig. 2, the classification logits output independently by the two branches are weighted and summed using the generated dynamic weights to obtain the network's final classification decision.

Prior to the DDF, the high-dimensional spatial and spectral feature maps generated by the preceding AGIB modules must be transformed into compact, high-level semantic tokens. Specifically, given the output feature tensor $X_{\mathrm{out}} \in \mathbb{R}^{C\times H\times W}$ from either the spatial or the spectral branch, we first flatten its spatial dimensions to obtain a 2-D sequence representation $X_{\mathrm{flat}} \in \mathbb{R}^{N\times C}$, where $N = H \times W$ denotes the total number of pixels. To aggregate these local pixel features into a predetermined number of semantic tokens (denoted as L), we design a learnable, attention-based tokenization mechanism. Let $W_{\mathrm{A}} \in \mathbb{R}^{L\times C}$ denote the learnable semantic mapping weights and $W_{\mathrm{V}} \in \mathbb{R}^{C\times d}$ represent the feature transformation weights, where d indicates the embedding dimension of the target tokens. The semantic tokenization process is formulated as follows:

$$A = \mathrm{Softmax}\left(X_{\mathrm{flat}} W_{\mathrm{A}}^{\mathrm{T}}\right) \tag{26}$$

$$V = X_{\mathrm{flat}} W_{\mathrm{V}} \tag{27}$$

$$F = A^{\mathrm{T}} V \tag{28}$$

where $A \in \mathbb{R}^{N\times L}$ denotes the semantic grouping attention map, indicating the contribution of each spatial pixel to the L semantic tokens, while $V \in \mathbb{R}^{N\times d}$ represents the projected feature values. The resulting token tensor $F \in \mathbb{R}^{L\times d}$ effectively compresses the complex spatial–spectral representations into a compact sequence space, thereby providing highly discriminative, high-level semantic features for the subsequent decision fusion. To comprehensively capture the global contextual information of the current pixel, we first concatenate the two features along the channel dimension to construct a fused feature representation

$$F_{\mathrm{cat}} = \mathrm{Concat}\left[F_{\mathrm{spe}}, F_{\mathrm{spa}}\right]. \tag{29}$$

Subsequently, feature F_{cat} is fed into the FFN to evaluate the decision confidence of the current pixel on the two branches. By learning the mapping relationship between deep semantic features and optimal fusion weights through end-to-end learning, the model output normalizes adaptive fusion weights for the two branches, providing a data-driven basis for subsequent decision-level fusion

$$\left[w_{\mathrm{spe}}, w_{\mathrm{spa}}\right] = \mathrm{Softmax}\left(\mathrm{FFN}\left(F_{\mathrm{cat}}\right)\right). \tag{30}$$

After the dual-branch spatial–spectral feature interaction and deep semantic feature purification are completed, the spectral branch and the spatial branch, respectively, output prediction logits corresponding to the land cover categories through mutually independent linear classifiers with nonshared weights. The dual-branch design is adopted to fully preserve the respective discriminative characteristics of the spectral and spatial branches. The two output streams, denoted as $P_{\mathrm{spe}} \in \mathbb{R}^{C}$ and $P_{\mathrm{spa}} \in \mathbb{R}^{C}$, respectively, reflect the independent classification discrimination results and confidence levels derived from the intrinsic spectral information and spatial context information, where C represents the number of land cover categories in the hyperspectral dataset. This provides the core basis for the subsequent DDF. The two sets of independently predicted logits were weighted and summed using dedicated confidence weights to obtain the final classification probability distribution. This design not only fully retains the respective discriminative characteristics of the spatial–spectral dual branches and avoids semantic conflict and information redundancy caused by the early fusion of heterogeneous spatial–spectral features but also dynamically adjusts the fusion strategy according to the spatial–spectral characteristics of each sample, realizing the DDF of spatial–spectral information

$$P_{\mathrm{final}} = w_{\mathrm{spe}} P_{\mathrm{spe}} + w_{\mathrm{spa}} P_{\mathrm{spa}}. \tag{31}$$

Algorithm 1 SFGANet for HSI Classification

Input: The original HSIs

Output: Predicted labels of all pixels

Initialization:

Batch size is 32; loss function is Cross-Entropy; number of channels after PCA is 30; patch size is 13 or 17; optimizer is Adam; learning rate is $1e$-4

Repeat:

1) Perform PCA to reduce dimension;
2) Perform two $3 \times 3 \times 3$ convolution layer;
3) Perform SBGM for band weighting;
4) Reshape the feature maps and perform DDFM to enhance features;
5) Extract spatial and spectral branch features using convolutional layers;
6) Perform AGIB for spatial and spectral feature fusion;
7) Tokenize spatial and spectral features using learned token parameters;
8) Generate spatial and spectral fusion weights;
9) Classify spatial and spectral tokens separately using linear layers;
10) Employ the DDF strategy to fuse the classification results from both branches and obtain the final output.

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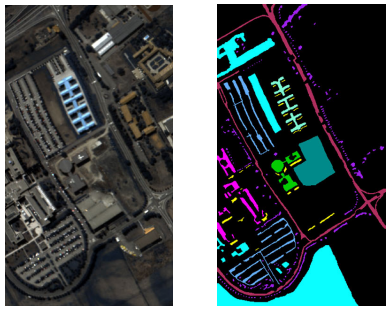


Fig. 6. PU dataset. (a) False-color map. (b) Ground-truth map.

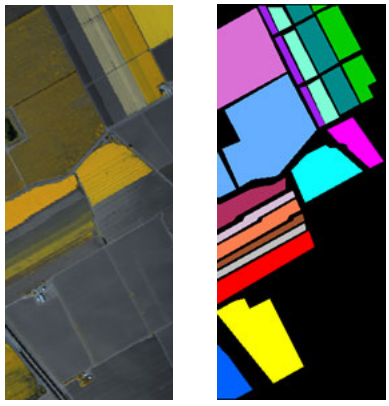


Fig. 7. Salinas dataset. (a) False-color map. (b) Ground-truth map.

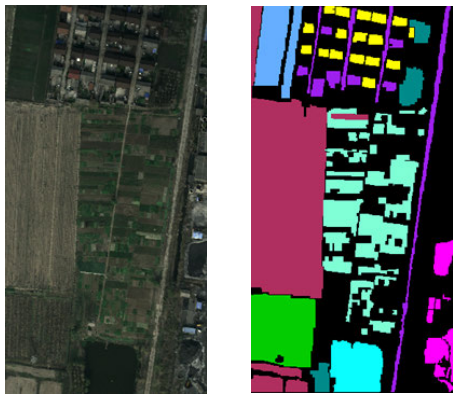


Fig. 8. Xuzhou dataset. (a) False-color map. (b) Ground-truth map.

SFGANet comprehensively optimized the entire workflow from feature extraction and cross-branch fusion to multipath decision-making and fully exploited the synergistic advantages of tokenization and dual-branch fusion. The highest value across all output categories corresponds to the final predicted label. The entire execution process of the proposed method is outlined in Algorithm 1.

IV. Experiments and Analysis

In this section, the effectiveness of the proposed method is verified via comprehensive experiments carried out on four publicly accessible datasets, and a comparative analysis is conducted between the proposed method and widely adopted comparative approaches. The overall arrangement of the experiments is structured as follows. First, the basic information and data properties of the four datasets employed in the experiments are described in detail. Next, the detailed experimental configurations are specified, and the value settings and influence mechanisms of key parameters are analyzed. Subsequently, the classification performance of the proposed method and the comparative methods was presented, and an in-depth analysis from both quantitative and qualitative dimensions was conducted. Finally, the independent function and contribution of each core module in the framework are validated through ablation experiments, and the rationality of the proposed framework design is further clarified.

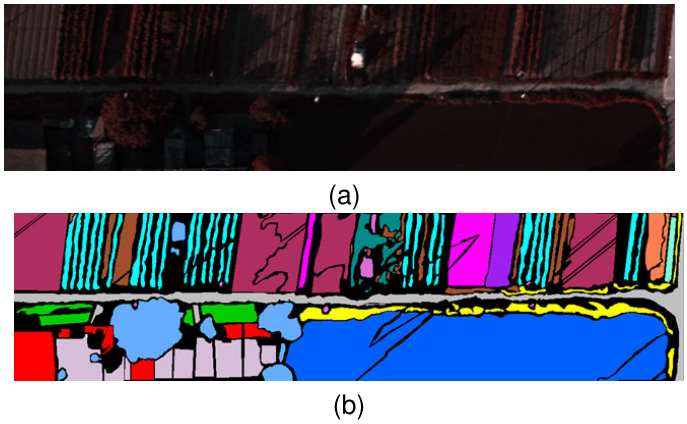


Fig. 9. WHU-Hi-HanChuan dataset. (a) False-color map. (b) Ground-truth map.

TABLE I
Details and Proportion of Samples Division of Four Datasets

Dataset Name	Spatial Resolution	Bands	Classes	Pixel Size	Training Ratio
Pavia University	1.3 m	103	9	610×340	0.5%
Salinas	3.7 m	204	16	512×217	0.5%
Xuzhou	0.73 m	436	9	500×260	0.5%
WHU-Hi-HanChuan	0.109 m	274	16	1217×303	0.5%

TABLE II
Training and Test Samples in the PU Dataset

ID	Land Cover Class	Color	Training	Test
1	Asphalt		33	6598
2	Meadows		93	18556
3	Gravel		10	2089
4	Trees		15	3049
5	Painted metal sheets		7	1338
6	Bare Soil		25	5004
7	Bitumen		7	1323
8	Self-Blocking Bricks		18	3664
9	Shadows		5	942
—	Total		213	42563

A. Dataset Description

Experiments were carried out on four publicly accessible HSI datasets, namely, Pavia University, Salinas, Xuzhou, and WHU-Hi-HanChuan. The detailed specifications of these datasets are summarized in Tables I–V, with corresponding

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TABLE III
TRAINING AND TEST SAMPLES IN THE SALINAS DATASET

ID	Land Cover Class	Color	Training	Test
1	Brocoli_green_weeds_1		10	1999
2	Brocoli_green_weeds_2		19	3707
3	Fallow		10	1966
4	Fallow_rough_plow		7	1387
5	Fallow_smooth		13	2665
6	Stubble		20	3939
7	Celery		18	3561
8	Grapes_untrained		56	11215
9	Soil_vinyard_develop		31	6172
10	Corn_senesced_green_weeds		16	3262
11	Lettuce_romaine_4wk		5	1063
12	Lettuce_romaine_5wk		10	1917
13	Lettuce_romaine_6wk		5	911
14	Lettuce_romaine_7wk		5	1065
15	Vinyard_untrained		36	7232
16	Vinyard_vertical_trellis		9	1798
—	Total		270	53859

TABLE IV
TRAINING AND TEST SAMPLES IN THE XUZHOU DATASET

ID	Land Cover Class	Color	Training	Test
1	Bareland_1		132	26264
2	Lakes		20	4007
3	Coals		14	2769
4	Cement		26	5188
5	Crop_1		66	13118
6	Trees		12	2424
7	Bareland_2		35	6955
8	Crop_2		24	4753
9	Red_tiles		15	3055
—	Total		344	68533

TABLE V
TRAINING AND TEST SAMPLES IN THE WHU-HI-HANCHUAN DATASET

ID	Land Cover Class	Color	Training	Test
1	Strawberry		224	44511
2	Cowpea		114	22639
3	Soybean		51	10236
4	Sorghum		27	5326
5	Water-spinach		6	1194
6	Watermelon		23	4510
7	Greens		29	5874
8	Trees_untrained		90	17888
9	Grass		47	9422
10	Red-roof		52	10464
11	Gray-roof		84	16827
12	Plastic		18	3661
13	Bare-soil		46	9070
14	Roads		93	18467
15	Bright-object		6	1130
16	Water		377	75024
—	Total		1287	256243

false-color composite maps and ground-truth maps provided in Figs. 6–9.

B. Experimental Setup

1) Evaluation Metrics: To measure the performance of the proposed method and other comparative methods, we use regular classification evaluation metrics for quantitative analysis, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (k). Larger values of each metric indicate a better classification performance.

2) Comparison Methods: To substantiate the efficacy of the proposed classification method, based on CNNs and transformer architectures, a comparative analysis was conducted with several representative methods. Among the CNN-based methods are 3D-CNN [62], DFFN [63], HybridSN [64], RSSAN [45], AB-LSTM [65], and SSEFN [66]; and Transformer-based methods include SF [47], GAHT [67], and methods based on the mixture of CNN and Transformer, including SSFTT [52], DCTN [56], and CSCANet [68].

3) Implementation Details: SFGANet and all comparative methods were implemented in the PyTorch [69] framework and executed on the same hardware platform equipped with an Intel I7-13700KF processor and an NVIDIA GeForce RTX 4090 graphics card. The training and test procedures of SFGANet were conducted using the Adam [70] optimizer with a learning rate of 0.0001. The batch size was set to 32, the training epochs to 100, and the PCA dimension to 30. All experiments were performed ten times, and the average of the ten runs was adopted as the final result. The parameter configurations of the comparative methods strictly followed the settings reported in their original literature, which ensures their performance attains the optimal level and guarantees experimental fairness.

C. Parameter Analysis

To comprehensively evaluate the proposed SFGANet model, we systematically analyzed several key hyperparameters for HSI classification: PCA dimensions, patch size, batch size, learning rate, token length, and the number of attention heads in GSSA. The size of these hyperparameters directly affects the complexity of the model and has an important impact on the classification performance of the model. When analyzing each hyperparameter, the other experimental parameters are set according to Section IV-B.

1) PCA Dimensions: To explore the influence of PCA-reduced dimensionality on classification performance, we performed variable-controlled experiments with five PCA dimensions (10, 20, 30, 40, and 50), with original high-dimensional data as the control group. As illustrated in Fig. 10(a), original data yielded the lowest OA on all datasets, while different datasets showed distinct accuracy trends after PCA processing. To balance classification performance and computational efficiency, we fixed the PCA dimension at 30 for all follow-up experiments.

2) Patch Size: To explore the impact of patch size on classification performance, we performed controlled variable experiments with five candidate patch sizes (9, 11, 13, 15, and 17) while fixing other parameters. As shown in Fig. 10(b), the optimal patch size for the PU dataset was 13, while the classification accuracy of the Salinas, Xuzhou, and WHU-Hi-HanChuan datasets fluctuated continuously with the increase of patch size. To balance classification performance and computational efficiency, we fixed the patch size to 17 for the

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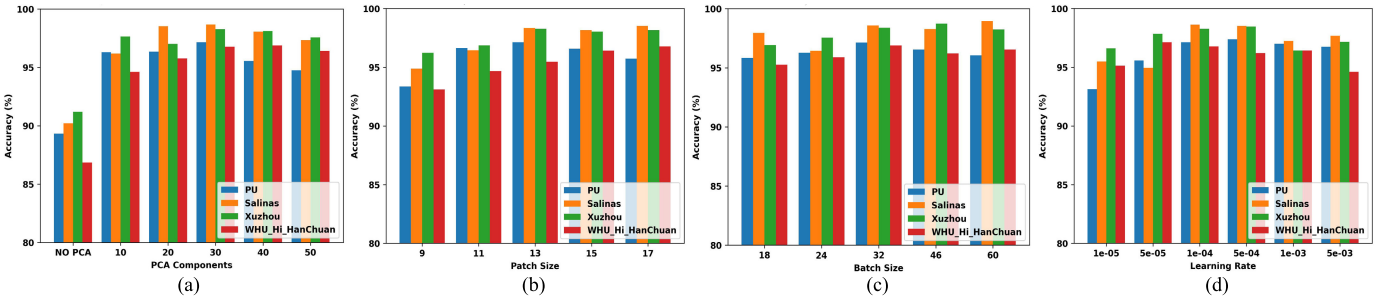


Fig. 10. Impact of different parameters on OA. (a) PCA dimensions. (b) Patch size. (c) Batch size. (d) Learning rate.

TABLE VI

CLASSIFICATION RESULTS FOR DIFFERENT METHODS ON THE PU DATASET WITH 0.5% TRAINING SAMPLES. IN THE TABLE, NO. INDICATES CLASS NUMBER, AND THE BEST RESULTS ARE IN BOLD

NO.	3D-CNN	DFFN	HybridSN	RSSAN	AB-LSTM	SSEFN	SF	SSFTT	GAHT	DCTN	CSCANet	SFGANet
1	93.91 ± 02.92	91.82 ± 03.88	91.59 ± 03.07	92.24 ± 02.48	89.17 ± 05.01	95.40 ± 01.06	81.71 ± 05.04	94.64 ± 03.08	94.88 ± 01.67	91.65 ± 00.77	96.76 ± 02.16	**97.65 ± 01.29**
2	96.02 ± 02.51	98.76 ± 00.90	98.66 ± 00.99	98.32 ± 00.82	94.38 ± 03.04	97.27 ± 00.65	94.85 ± 02.81	99.31 ± 00.88	99.43 ± 00.34	91.77 ± 00.30	**99.82 ± 00.25**	99.79 ± 00.15
3	54.63 ± 16.24	84.58 ± 07.65	82.32 ± 07.84	62.99 ± 14.63	47.67 ± 08.25	**90.17 ± 05.00**	55.03 ± 15.53	76.63 ± 11.32	82.44 ± 09.26	72.94 ± 01.74	81.33 ± 04.86	83.16 ± 08.55
4	93.62 ± 01.83	72.69 ± 05.78	73.09 ± 06.05	93.03 ± 02.19	89.01 ± 03.57	**99.36 ± 00.79**	85.95 ± 06.32	94.60 ± 03.35	95.20 ± 02.03	94.12 ± 00.22	90.55 ± 04.21	93.18 ± 01.73
5	98.31 ± 03.31	95.39 ± 04.54	92.30 ± 07.94	96.14 ± 03.41	96.34 ± 02.24	99.45 ± 00.95	99.92 ± 00.09	99.67 ± 00.67	**100.00 ± 00.00**	98.96 ± 00.20	99.93 ± 00.22	99.25 ± 00.80
6	71.58 ± 10.11	96.51 ± 02.59	96.78 ± 02.49	85.88 ± 05.38	70.39 ± 11.30	97.00 ± 00.45	67.82 ± 09.02	94.16 ± 05.90	95.58 ± 02.56	97.43 ± 01.21	95.58 ± 02.81	**99.21 ± 01.16**
7	65.63 ± 14.17	91.76 ± 06.82	91.43 ± 05.72	73.24 ± 07.16	66.89 ± 11.51	98.44 ± 01.17	57.51 ± 08.53	81.54 ± 10.45	88.93 ± 06.79	92.42 ± 01.74	99.36 ± 00.66	**99.74 ± 00.72**
8	88.90 ± 05.70	**95.77 ± 01.31**	95.45 ± 01.72	83.94 ± 06.40	76.79 ± 08.14	86.87 ± 03.18	70.61 ± 09.53	92.91 ± 04.57	90.95 ± 08.63	86.55 ± 01.61	89.17 ± 04.10	92.39 ± 03.31
9	98.98 ± 00.99	64.46 ± 05.45	67.33 ± 05.59	94.60 ± 05.54	98.60 ± 01.09	92.31 ± 00.49	93.16 ± 04.42	93.51 ± 07.03	**99.41 ± 00.60**	96.60 ± 00.58	87.03 ± 07.15	93.51 ± 03.94
OA(%)	89.19 ± 02.03	93.52 ± 00.47	92.57 ± 00.55	91.63 ± 01.03	85.86 ± 01.61	96.35 ± 00.50	83.92 ± 01.97	95.31 ± 01.59	96.09 ± 00.56	88.08 ± 00.49	96.08 ± 00.76	**97.13 ± 00.27**
AA(%)	84.62 ± 02.85	87.97 ± 01.04	87.66 ± 01.34	86.71 ± 02.31	81.03 ± 02.01	95.14 ± 00.70	78.51 ± 02.43	91.89 ± 02.13	94.09 ± 00.71	92.40 ± 00.83	93.28 ± 01.10	**95.32 ± 00.76**
Kappa	85.53 ± 02.73	91.39 ± 00.63	91.13 ± 00.75	88.84 ± 01.40	81.13 ± 02.12	95.14 ± 00.67	78.50 ± 02.59	93.77 ± 02.14	94.82 ± 00.73	84.72 ± 00.65	94.77 ± 01.01	**95.91 ± 00.36**
Training Time(s)	25.52 ± 02.84	106.32 ± 09.70	26.17 ± 04.83	74.26 ± 05.73	107.29 ± 07.33	82.80 ± 02.59	187.66 ± 11.41	25.26 ± 00.72	66.13 ± 01.05	284.93 ± 06.40	18.37 ± 01.78	30.28 ± 01.36
Test Time(s)	08.84 ± 00.57	67.44 ± 04.17	07.26 ± 01.16	20.85 ± 01.79	19.82 ± 01.19	63.89 ± 02.23	58.67 ± 04.22	34.67 ± 00.62	27.28 ± 00.39	238.86 ± 03.21	08.11 ± 01.69	17.46 ± 01.24

Salinas, Xuzhou, and WHU-Hi-HanChuan datasets and 13 for the PU dataset in all subsequent experiments.

3) Batch Size: To explore the influence of batch size on classification performance, we performed variable-controlled experiments with five batch sizes (18, 24, 32, 46, and 60) while keeping all other parameters fixed. As shown in Fig. 10(c), the PU and WHU-Hi-HanChuan datasets achieved optimal performance at a batch size of 32, while the classification accuracy of the Salinas and Xuzhou datasets fluctuated within a narrow range. To balance classification performance, computational efficiency, and generalization ability, we fixed the batch size at 32 for all subsequent experiments.

4) Learning Rate: To explore the impact of learning rate on model performance and convergence, we performed controlled variable experiments with six candidate values (1e-5, 5e-5, 1e-4, 5e-4, 1e-3, and 5e-3). As shown in Fig. 10(d), raising the learning rate accelerated convergence and improved accuracy up to a critical threshold, beyond which performance dropped. Extremely low rates led to slow convergence, while excessively high values prevented optimal convergence. To balance efficiency and performance, a learning rate of 1e-4 was adopted for all subsequent experiments.

5) Impact of Token Length: The number of tokens from AGIB's postmodule dual-branch tokenization directly affects spatial–spectral feature abstraction granularity and model computational efficiency. A smaller token count reduces feature representation richness, while a larger one notably increases tokenization-stage computational overhead and parameters. Fig. 11(a) presents OA across four datasets under different token counts. The optimal balance between classification accuracy and inference efficiency was achieved with a token count of 24.

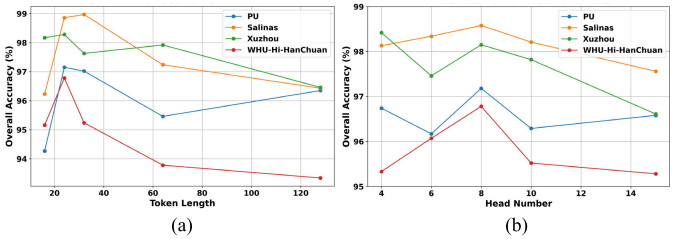


Fig. 11. OA of the proposed SFGANet under different token lengths and attention head numbers on four hyperspectral datasets. (a) Illustrates the overall accuracy (OA) curves of the proposed SFGANet model on four hyperspectral datasets under different token length settings, which quantitatively analyzes how the number of learnable semantic tokens affects classification performance and computational efficiency to select the optimal token count. (b) Presents the OA variation of SFGANet with varying numbers of attention heads inside the GSSA module across the four datasets, exploring the trade-off between the feature representation capacity of multi-head geometric semantic synergistic attention and model computational complexity to determine the most suitable head number for all experimental datasets.

6) Impact of the Number of Attention Heads in GSSA: The number of attention heads in the GSSA module directly affects the model's representation capability and computational complexity. To determine the optimal head count, experiments were conducted with candidate values of 4, 6, 8, 10, and 15. Fig. 11(b) presents the OA of the four datasets under different attention head numbers. The best classification performance was achieved with eight heads for the PU, Salinas, and WHU-Hi-HanChuan datasets and four heads for the Xuzhou dataset. For consistency, the attention head number was uniformly set to eight across all four datasets.

D. Classification Results and Analysis

1) Quantitative Results and Analysis: The per-class accuracy, OA, AA, and K of the proposed method and the comparison methods across four datasets are presented in

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TABLE VII

CLASSIFICATION RESULTS FOR DIFFERENT METHODS ON THE SALINAS DATASET WITH 0.5% TRAINING SAMPLES. IN THE TABLE, NO. INDICATES CLASS NUMBER, AND THE BEST RESULTS ARE IN BOLD

NO.	3D-CNN	DFFN	HybridSN	RSSAN	AB-LSTM	SSEFN	SF	SSFTT	GAHT	DCTN	CSCANet	SFGANet
1	96.84 ± 02.57	99.60 ± 00.43	94.58 ± 06.80	96.03 ± 04.01	94.24 ± 04.58	**100.00 ± 00.00**	94.22 ± 01.68	99.22 ± 01.49	97.60 ± 04.16	94.33 ± 00.02	99.81 ± 00.54	99.58 ± 01.18
2	98.35 ± 04.74	98.34 ± 02.10	96.64 ± 04.86	99.14 ± 00.99	98.77 ± 00.85	99.81 ± 00.32	99.02 ± 00.63	98.52 ± 03.38	**100.00 ± 00.00**	**100.00 ± 00.00**	**100.00 ± 00.00**	**100.00 ± 00.00**
3	86.57 ± 08.43	95.55 ± 04.41	90.46 ± 10.47	95.33 ± 02.47	87.54 ± 07.96	99.93 ± 00.09	92.47 ± 04.01	87.33 ± 10.79	99.39 ± 00.33	97.88 ± 00.37	**100.00 ± 00.00**	**100.00 ± 00.00**
4	98.30 ± 00.90	96.25 ± 02.21	90.36 ± 15.65	98.22 ± 02.05	98.10 ± 02.41	**98.57 ± 01.40**	95.44 ± 01.77	97.21 ± 03.55	98.56 ± 01.47	97.90 ± 00.08	96.26 ± 01.77	96.47 ± 02.06
5	91.27 ± 09.08	93.85 ± 03.40	81.75 ± 02.70	95.19 ± 03.98	96.39 ± 01.88	99.01 ± 00.91	86.18 ± 05.18	93.26 ± 11.07	97.00 ± 02.70	97.70 ± 00.28	**99.17 ± 01.15**	96.97 ± 03.34
6	99.48 ± 00.50	99.06 ± 02.09	98.93 ± 01.39	99.38 ± 01.24	97.89 ± 01.86	**100.00 ± 00.00**	99.58 ± 00.31	99.58 ± 00.53	**100.00 ± 00.00**	97.60 ± 00.02	98.92 ± 01.77	99.06 ± 01.48
7	99.40 ± 00.38	98.89 ± 01.50	97.85 ± 02.76	97.20 ± 01.79	98.67 ± 00.96	**99.99 ± 00.02**	97.91 ± 01.36	99.48 ± 01.07	99.90 ± 00.07	96.28 ± 00.22	99.92 ± 00.16	99.87 ± 00.20
8	80.72 ± 07.69	93.98 ± 02.28	90.65 ± 04.99	80.27 ± 06.07	80.16 ± 03.39	94.22 ± 02.74	83.47 ± 02.75	86.12 ± 02.52	89.42 ± 03.62	92.44 ± 00.56	96.68 ± 01.35	**97.30 ± 01.21**
9	98.05 ± 01.52	99.05 ± 01.15	99.18 ± 01.19	99.16 ± 00.77	97.78 ± 01.68	99.41 ± 00.10	95.35 ± 02.94	99.51 ± 00.74	99.96 ± 00.08	99.37 ± 00.02	**100.00 ± 00.00**	**100.00 ± 00.00**
10	85.48 ± 07.05	97.36 ± 03.06	96.28 ± 02.72	91.43 ± 05.14	77.51 ± 05.74	**98.53 ± 00.77**	85.42 ± 03.83	95.40 ± 02.61	95.82 ± 02.41	95.80 ± 00.26	97.69 ± 01.91	97.87 ± 01.37
11	82.26 ± 06.97	97.11 ± 01.44	96.20 ± 02.93	80.93 ± 07.62	65.21 ± 28.28	97.37 ± 01.64	80.17 ± 06.76	90.98 ± 11.34	94.04 ± 05.80	94.05 ± 00.60	98.29 ± 01.81	**99.87 ± 00.14**
12	98.26 ± 02.41	98.65 ± 01.47	95.20 ± 03.86	98.29 ± 01.55	96.14 ± 02.40	98.57 ± 01.08	95.95 ± 04.90	97.80 ± 03.71	**99.86 ± 00.18**	96.55 ± 00.03	99.47 ± 00.77	99.09 ± 01.33
13	99.21 ± 00.77	92.90 ± 05.18	91.14 ± 08.13	97.39 ± 02.78	94.40 ± 06.48	96.92 ± 03.52	94.43 ± 06.13	**99.86 ± 00.43**	99.43 ± 00.80	96.97 ± 00.07	90.78 ± 06.27	93.27 ± 07.79
14	96.15 ± 00.87	95.79 ± 04.17	95.64 ± 04.00	92.47 ± 03.26	84.48 ± 08.79	93.59 ± 03.27	94.54 ± 04.46	**98.22 ± 01.42**	98.19 ± 00.74	96.47 ± 00.21	93.86 ± 08.98	93.89 ± 07.16
15	73.87 ± 06.88	93.90 ± 03.73	91.00 ± 06.58	77.99 ± 07.18	64.50 ± 18.02	93.81 ± 01.72	72.12 ± 06.58	92.06 ± 03.08	86.07 ± 05.95	85.51 ± 00.98	96.81 ± 00.97	**98.74 ± 00.99**
16	80.14 ± 10.79	85.11 ± 06.11	84.39 ± 05.96	90.76 ± 05.53	80.99 ± 13.62	**100.00 ± 00.00**	82.82 ± 09.43	86.66 ± 12.96	93.82 ± 04.30	75.04 ± 00.74	99.54 ± 00.60	99.45 ± 00.34
OA(%)	88.94 ± 01.07	96.02 ± 00.83	93.37 ± 01.46	90.49 ± 00.83	86.40 ± 02.23	97.38 ± 00.54	88.56 ± 01.21	93.91 ± 00.78	94.99 ± 00.40	91.67 ± 00.24	98.14 ± 00.42	**98.86 ± 00.36**
AA(%)	91.52 ± 01.14	95.96 ± 00.79	93.14 ± 01.78	93.07 ± 00.82	88.30 ± 02.24	98.11 ± 00.35	90.57 ± 01.64	95.08 ± 01.14	96.82 ± 00.35	98.07 ± 00.18	97.95 ± 00.64	**98.21 ± 00.44**
Kappa	87.68 ± 01.18	95.57 ± 00.93	92.62 ± 01.62	89.42 ± 00.92	84.84 ± 02.51	97.08 ± 00.60	87.26 ± 01.35	93.22 ± 00.87	94.42 ± 00.45	90.78 ± 00.27	98.34 ± 00.46	**98.35 ± 00.40**
Training Time(s)	40.77 ± 00.57	131.97 ± 05.84	32.72 ± 02.98	100.70 ± 03.31	178.24 ± 01.93	140.19 ± 02.06	263.46 ± 15.81	41.03 ± 03.15	102.59 ± 00.70	327.48 ± 03.79	26.76 ± 01.50	38.21 ± 01.64
Test Time(s)	09.89 ± 00.44	63.09 ± 03.54	08.48 ± 01.26	14.28 ± 00.39	15.41 ± 00.40	80.64 ± 02.56	40.21 ± 00.78	24.51 ± 01.79	18.75 ± 00.56	125.99 ± 02.58	11.34 ± 00.90	22.63 ± 02.37

TABLE VIII

CLASSIFICATION RESULTS FOR DIFFERENT METHODS ON THE XUZHOU DATASET WITH 0.5% TRAINING SAMPLES. IN THE TABLE, NO. INDICATES CLASS NUMBER, AND THE BEST RESULTS ARE IN BOLD

NO.	3D-CNN	DFFN	HybridSN	RSSAN	AB-LSTM	SSEFN	SF	SSFTT	GAHT	DCTN	CSCANet	SFGANet
1	96.89 ± 00.90	95.71 ± 01.54	95.96 ± 01.06	96.72 ± 00.73	91.85 ± 01.86	**99.76 ± 00.18**	95.25 ± 01.47	97.50 ± 00.65	97.98 ± 00.63	92.86 ± 00.07	97.13 ± 00.97	97.31 ± 00.41
2	97.00 ± 01.15	98.89 ± 00.85	98.54 ± 01.41	96.33 ± 03.08	97.35 ± 01.45	**99.66 ± 00.92**	93.84 ± 02.03	98.39 ± 01.36	99.50 ± 00.48	93.64 ± 00.46	99.28 ± 00.62	99.27 ± 00.64
3	86.13 ± 06.35	91.71 ± 08.31	89.38 ± 05.45	87.94 ± 04.17	65.92 ± 19.87	95.92 ± 01.78	85.60 ± 05.54	90.57 ± 03.92	95.35 ± 04.50	82.33 ± 00.66	**98.24 ± 01.37**	97.95 ± 01.49
4	88.16 ± 03.03	86.91 ± 04.71	85.52 ± 03.98	90.68 ± 01.64	85.92 ± 03.59	97.45 ± 02.08	86.33 ± 02.62	93.95 ± 02.39	94.11 ± 02.88	95.44 ± 00.36	96.83 ± 03.53	**97.48 ± 02.00**
5	96.33 ± 00.97	97.00 ± 01.76	97.23 ± 00.87	97.52 ± 01.31	87.35 ± 02.00	96.45 ± 01.23	95.45 ± 01.29	97.57 ± 01.16	98.90 ± 00.70	99.01 ± 00.06	98.51 ± 01.00	**99.19 ± 00.48**
6	83.03 ± 05.08	94.43 ± 05.76	95.94 ± 02.51	86.71 ± 07.71	63.29 ± 07.94	95.63 ± 02.10	81.58 ± 04.66	91.50 ± 08.98	96.55 ± 01.54	95.87 ± 00.29	**98.55 ± 00.93**	98.08 ± 01.97
7	92.53 ± 04.89	97.13 ± 02.15	96.87 ± 02.30	93.88 ± 03.70	83.07 ± 07.85	93.12 ± 02.03	93.02 ± 02.73	94.98 ± 04.88	97.09 ± 01.98	93.70 ± 00.17	99.06 ± 00.56	**99.68 ± 00.25**
8	97.21 ± 01.18	99.12 ± 00.80	98.94 ± 00.93	97.26 ± 01.38	93.82 ± 02.89	97.74 ± 01.60	94.70 ± 01.19	97.01 ± 03.94	98.60 ± 01.00	94.28 ± 00.48	98.87 ± 01.04	**99.31 ± 01.08**
9	93.89 ± 02.66	88.63 ± 04.27	89.44 ± 05.89	95.74 ± 02.86	78.45 ± 12.58	98.13 ± 01.69	92.15 ± 01.96	97.48 ± 01.37	**98.64 ± 00.87**	95.19 ± 00.47	94.59 ± 02.70	96.99 ± 02.01
OA(%)	94.65 ± 00.51	95.34 ± 00.80	95.23 ± 00.72	95.39 ± 00.63	87.45 ± 01.81	97.67 ± 00.40	93.26 ± 00.67	96.51 ± 00.90	97.78 ± 00.28	90.64 ± 00.06	97.80 ± 00.51	**98.28 ± 00.29**
AA(%)	92.35 ± 01.09	94.39 ± 01.36	94.20 ± 00.99	93.64 ± 01.09	83.00 ± 03.31	97.10 ± 00.67	90.88 ± 00.82	95.44 ± 01.65	97.41 ± 00.48	97.68 ± 00.19	97.89 ± 00.40	**98.35 ± 00.42**
Kappa	93.20 ± 00.66	94.10 ± 01.01	94.05 ± 00.91	94.15 ± 00.79	84.13 ± 02.27	97.05 ± 00.51	91.45 ± 00.84	95.57 ± 01.14	97.18 ± 00.36	88.40 ± 00.08	97.21 ± 00.65	**97.73 ± 00.36**
Training Time(s)	69.12 ± 02.00	212.58 ± 14.72	49.17 ± 02.39	161.00 ± 02.60	378.94 ± 22.79	186.284 ± 03.923	531.87 ± 08.56	81.78 ± 06.88	143.65 ± 57.02	624.39 ± 08.70	47.01 ± 02.65	51.88 ± 02.29
Test Time(s)	19.10 ± 00.95	146.94 ± 29.22	13.41 ± 01.67	24.26 ± 00.51	31.90 ± 02.41	100.285 ± 07.353	88.58 ± 03.55	80.32 ± 04.28	24.50 ± 09.68	193.61 ± 03.55	12.97 ± 00.83	28.80 ± 00.98

TABLE IX

CLASSIFICATION RESULTS FOR DIFFERENT METHODS ON THE WHU-HI-HANCHUAN DATASET WITH 0.5% TRAINING SAMPLES. IN THE TABLE, NO. INDICATES CLASS NUMBER, AND THE BEST RESULTS ARE IN BOLD

NO.	3D-CNN	DFFN	HybridSN	RSSAN	AB-LSTM	SSEFN	SF	SSFTT	GAHT	DCTN	CSCANet	SFGANet
1	94.83 ± 01.80	**98.82 ± 00.80**	98.21 ± 01.55	95.97 ± 01.32	94.62 ± 00.64	92.52 ± 01.08	94.00 ± 00.56	95.13 ± 03.12	97.37 ± 00.48	94.61 ± 00.16	98.81 ± 00.40	98.78 ± 00.84
2	85.62 ± 02.30	96.41 ± 02.29	88.13 ± 04.32	87.41 ± 02.22	81.47 ± 01.93	93.43 ± 01.68	76.16 ± 03.29	91.48 ± 01.92	92.94 ± 02.20	94.88 ± 00.19	93.72 ± 03.78	**97.71 ± 00.89**
3	76.91 ± 08.89	**96.98 ± 02.45**	87.24 ± 06.62	86.98 ± 03.53	74.53 ± 05.18	94.18 ± 01.99	88.35 ± 04.03	89.82 ± 05.37	94.78 ± 02.33	89.91 ± 01.57	96.82 ± 02.53	96.86 ± 02.54
4	91.92 ± 03.37	97.46 ± 02.51	95.35 ± 03.54	95.25 ± 01.59	88.85 ± 06.31	95.78 ± 04.09	94.50 ± 02.13	93.01 ± 04.57	97.33 ± 01.59	96.67 ± 00.17	98.78 ± 01.15	**99.10 ± 00.98**
5	56.17 ± 15.93	93.73 ± 04.26	78.07 ± 16.23	53.78 ± 11.69	24.90 ± 08.81	**94.14 ± 04.20**	39.68 ± 15.08	67.79 ± 16.63	73.47 ± 11.72	90.25 ± 03.86	87.21 ± 05.73	91.73 ± 06.96
6	22.85 ± 11.08	65.90 ± 12.42	42.34 ± 22.04	34.12 ± 04.97	20.52 ± 10.31	81.91 ± 03.90	27.44 ± 04.33	50.96 ± 16.24	61.57 ± 08.15	73.73 ± 02.39	76.52 ± 07.50	**82.25 ± 07.62**
7	79.76 ± 06.45	91.39 ± 03.50	78.95 ± 12.21	79.91 ± 12.82	76.00 ± 04.73	77.42 ± 04.74	75.28 ± 06.66	78.09 ± 10.94	83.95 ± 06.26	92.87 ± 00.38	93.03 ± 03.26	**95.34 ± 02.61**
8	77.64 ± 06.77	87.63 ± 03.18	76.28 ± 07.91	73.60 ± 04.14	69.41 ± 03.93	92.31 ± 02.32	72.09 ± 02.84	85.44 ± 04.22	85.43 ± 03.77	91.17 ± 01.00	89.68 ± 11.69	**93.11 ± 02.36**
9	72.14 ± 08.35	88.68 ± 04.06	77.51 ± 06.82	71.46 ± 07.16	59.41 ± 05.79	85.84 ± 01.46	62.07 ± 04.37	81.56 ± 07.02	87.06 ± 05.54	**91.72 ± 00.56**	90.07 ± 02.94	91.11 ± 03.72
10	87.90 ± 03.69	95.48 ± 05.13	88.60 ± 08.32	95.13 ± 02.02	92.08 ± 02.26	**99.32 ± 00.34**	94.25 ± 01.41	94.31 ± 06.79	97.40 ± 01.20	90.68 ± 00.42	97.99 ± 00.96	97.36 ± 01.30
11	92.79 ± 03.81	97.67 ± 00.73	92.44 ± 06.67	93.15 ± 03.61	92.56 ± 02.52	91.82 ± 01.59	92.81 ± 02.07	95.40 ± 03.12	93.95 ± 02.60	91.83 ± 00.48	97.62 ± 01.35	**98.04 ± 01.43**
12	34.32 ± 09.12	90.18 ± 07.85	69.95 ± 14.27	61.41 ± 12.05	26.18 ± 10.74	91.98 ± 03.65	49.01 ± 06.38	72.45 ± 11.34	73.52 ± 08.59	81.42 ± 01.59	92.48 ± 03.24	**93.61 ± 05.15**
13	55.52 ± 08.03	82.01 ± 05.46	54.33 ± 16.41	67.88 ± 06.64	54.26 ± 04.82	**87.15 ± 02.88**	60.81 ± 04.98	75.33 ± 06.06	75.39 ± 06.12	84.52 ± 00.84	82.45 ± 03.47	81.61 ± 03.67
14	82.95 ± 03.11	93.05 ± 02.19	83.22 ± 05.63	86.15 ± 03.29	82.93 ± 02.25	90.19 ± 02.41	83.69 ± 01.25	89.50 ± 04.26	90.86 ± 01.18	88.47 ± 00.48	94.76 ± 01.66	**96.14 ± 01.06**
15	73.38 ± 10.02	69.87 ± 09.56	63.75 ± 09.91	60.45 ± 10.42	59.79 ± 10.87	**92.51 ± 10.93**	58.25 ± 08.39	76.80 ± 08.65	68.68 ± 17.22	79.50 ± 04.75	75.58 ± 09.27	78.70 ± 10.38
16	99.05 ± 00.83	99.57 ± 00.27	99.37 ± 00.77	99.21 ± 00.41	99.02 ± 00.39	98.94 ± 00.59	98.55 ± 00.32	98.89 ± 01.06	99.57 ± 00.23	96.64 ± 00.06	99.80 ± 00.17	**99.86 ± 00.10**
OA(%)	87.03 ± 01.06	94.33 ± 00.91	89.39 ± 01.60	89.10 ± 00.50	85.07 ± 00.92	93.64 ± 00.29	86.20 ± 00.38	91.65 ± 00.91	93.32 ± 00.45	90.31 ± 00.20	95.52 ± 01.24	**96.78 ± 00.28**
AA(%)	73.98 ± 02.45	90.30 ± 02.09	79.61 ± 03.51	77.62 ± 01.51	68.53 ± 02.19	91.21 ± 00.90	72.93 ± 01.18	83.50 ± 02.42	85.83 ± 01.86	90.13 ± 00.36	91.58 ± 01.36	**93.21 ± 01.03**
Kappa	84.78 ± 01.23	94.53 ± 01.06	87.57 ± 01.88	87.23 ± 00.59	82.50 ± 01.08	92.54 ± 00.34	83.85 ± 00.44	90.23 ± 01.07	92.17 ± 00.53	88.77 ± 00.23	95.15 ± 01.45	**96.23 ± 00.34**
Training Time(s)	197.30 ± 01.29	776.27 ± 12.68	168.04 ± 08.04	516.25 ± 13.46	976.34 ± 68.38	559.97 ± 36.24	1423.21 ± 21.43	245.67 ± 13.55	519.02 ± 35.76	1727.98 ± 32.57	84.58 ± 21.32	194.76 ± 10.87
Test Time(s)	39.06 ± 01.99	283.80 ± 08.28	52.68 ± 06.53	57.66 ± 00.60	67.15 ± 05.17	267.75 ± 08.59	224.92 ± 06.71	116.36 ± 06.82	71.73 ± 06.28	435.81 ± 08.60	44.80 ± 12.51	116.02 ± 04.85

Tables VI–IX, respectively, with the best results highlighted in bold. Overall, the CNN-based method failed to effectively incorporate contextual information to establish global dependencies. Although the Transformer-based method has advantages in capturing long-range dependencies, it performs poorly in handling local fine-grained features. The hybrid method, which combines elements of both CNN and Transformer, integrates the strengths of both approaches to some extent; consequently, it achieved the highest category accuracy in most cases, directly demonstrating its advantage in capturing fine-grained category features. Furthermore, our method achieved the highest number of best category accuracy results, further validating its superiority.

In HSI classification tasks, a model's joint spatial–spectral feature modeling ability and dataset adaptability directly determine its performance. Experimental results show that, with the DDFM and AGIB strategies proposed in this work, SFGANet achieved outstanding performance across all four datasets, with OA values that surpassed the best-performing baselines by 0.78%, 0.72%, 0.48%, and 1.26%, respectively. On the PU dataset, SSEFN effectively mitigated overfitting in small-sample scenarios and achieved excellent classification performance via multiscale spatial–spectral collaborative enhancement and multibranch decision fusion. In contrast, SFGANet's frequency–spatial domain parallel lightweight feature extraction architecture avoids the massive

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Fig. 12. Classification maps for the PU dataset. (a) 3D-CNN, (b) DFFN, (c) HybridSN, (d) RSSAN, (e) AB-LSTM, (f) SSEFN, (g) SF, (h) SSFTT, (i) GAHT, (j) DCTN, (k) CSCANet, and (l) SFGANet.

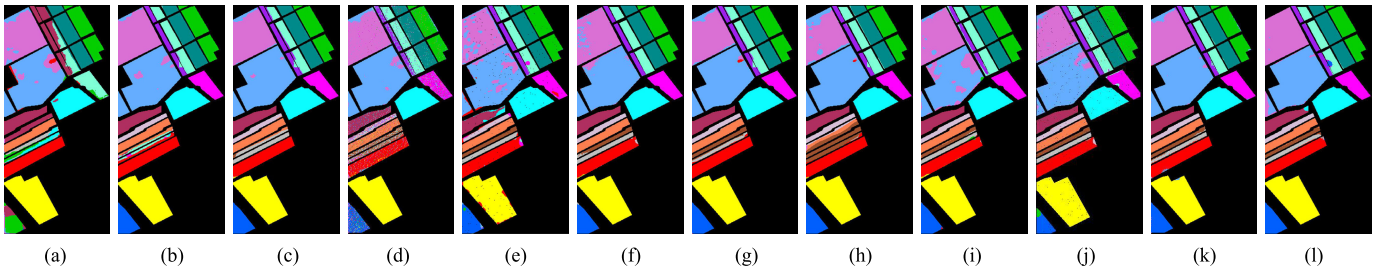


Fig. 13. Classification maps for the Salinas dataset. (a) 3D-CNN, (b) DFFN, (c) HybridSN, (d) RSSAN, (e) AB-LSTM, (f) SSEFN, (g) SF, (h) SSFTT, (i) GAHT, (j) DCTN, (k) CSCANet, and (l) SFGANet.

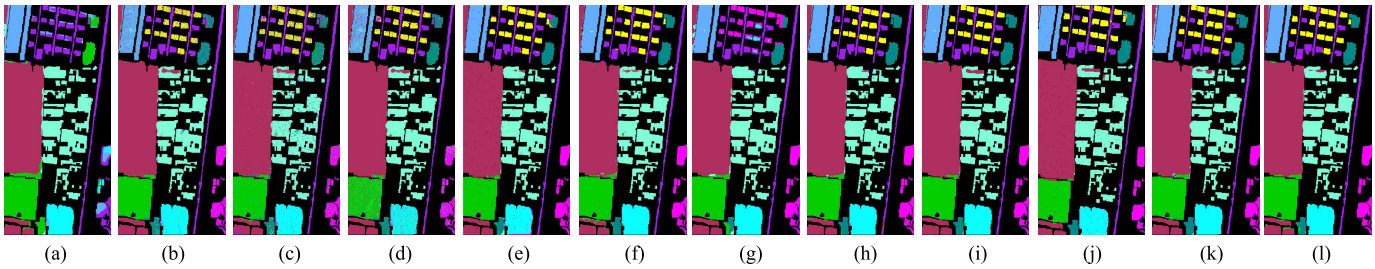


Fig. 14. Classification maps for the Xuzhou dataset. (a) 3D-CNN, (b) DFFN, (c) HybridSN, (d) RSSAN, (e) AB-LSTM, (f) SSEFN, (g) SF, (h) SSFTT, (i) GAHT, (j) DCTN, (k) CSCANet, and (l) SFGANet.

computational overhead of SSEFN's 20 parallel subnetworks; significantly shorter training and inference times were recorded in experiments, and a better tradeoff between accuracy and computational efficiency was achieved. On the Salinas dataset, CSCANet achieved extremely high accuracy for classes with continuous spectral features and flat spatial distribution, such as Fallow_smooth, via cascaded spatial cross-attention and dual-branch spatial–spectral feature separation. On the Xuzhou dataset, GAHT delivered good performance for man-made classes with strong spectral discriminability and regular spatial textures, such as Red_tiles, via grouped pixel embedding and hierarchical Transformer global–local collaborative modeling. Therefore, further optimization of the lightweight frequency–spatial parallel architecture and deep spatial–spectral feature fusion within the AGIB module is recommended to improve the classification performance of SFGANet. On the WHU-Hi-Hanchuan dataset, SFGANet leveraged its inherent strengths to address the challenges of the dataset's high-resolution UAV imagery, complex land cover types, large scale differences between crops and urban–rural features, and intraclass spectral heterogeneity and interclass spectral overlap. It effectively resolved overfitting and insufficient feature discriminability in small-sample classification on this dataset and outperformed all other comparative methods on the three core metrics: OA, AA, and K.

2) Visual Evaluation and Analysis: Figs. 12–15 present the classification results of SFGANet proposed in this work and various comparison methods across all four datasets. It can be seen that the classification map generated by SFGANet has minimal salt-and-pepper misclassification noise and superior spatial continuity of ground objects, with significantly better overall agreement with the ground truth than all other comparative methods. Taking the WHU-Hi-HanChuan dataset as an example, this complex urban–rural fringe scene has extensive shadowed areas caused by low solar elevation angle during image acquisition, where Gray-roof and Trees are typical confusable classes due to shadow-induced highly similar spectral features. For this challenge, SFGANet outperformed other methods significantly, effectively reducing interclass misclassification while maintaining clear class boundaries. SSEFN and CSCANet achieved the next-best performance but still slightly underperformed SFGANet in distinguishing subtle spectral features of the two classes and suppressing local misclassification in shadowed areas.

For further visual comparison of the performance of SFGANet with that of the comparison models, a comparative

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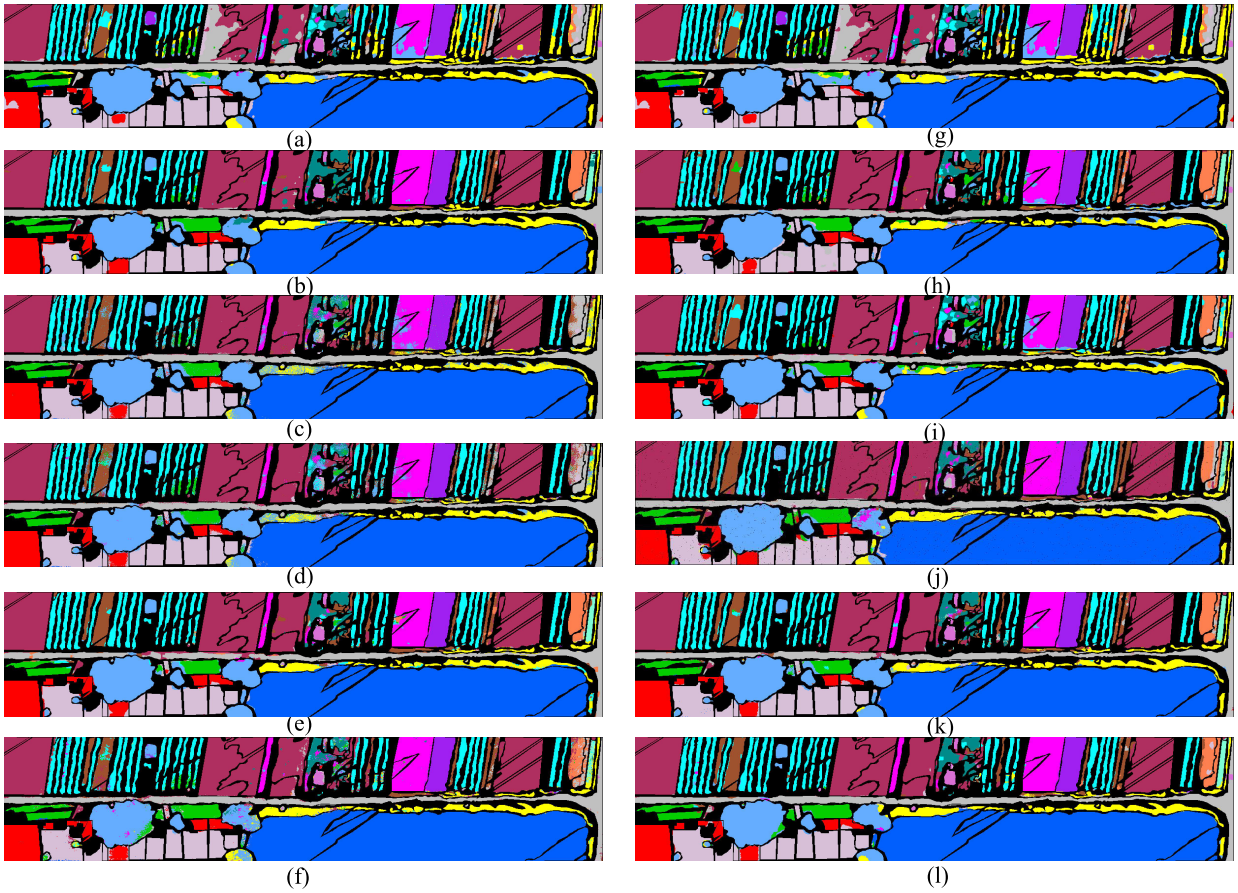


Fig. 15. Classification maps for the WHU-Hi-HanChuan dataset. (a) 3D-CNN, (b) DFFN, (c) HybridSN, (d) RSSAN, (e) AB-LSTM, (f) SSEFN, (g) SF, (h) SSFTT, (i) GAHT, (j) DCTN, (k) CSCANet, and (l) SFGANet.

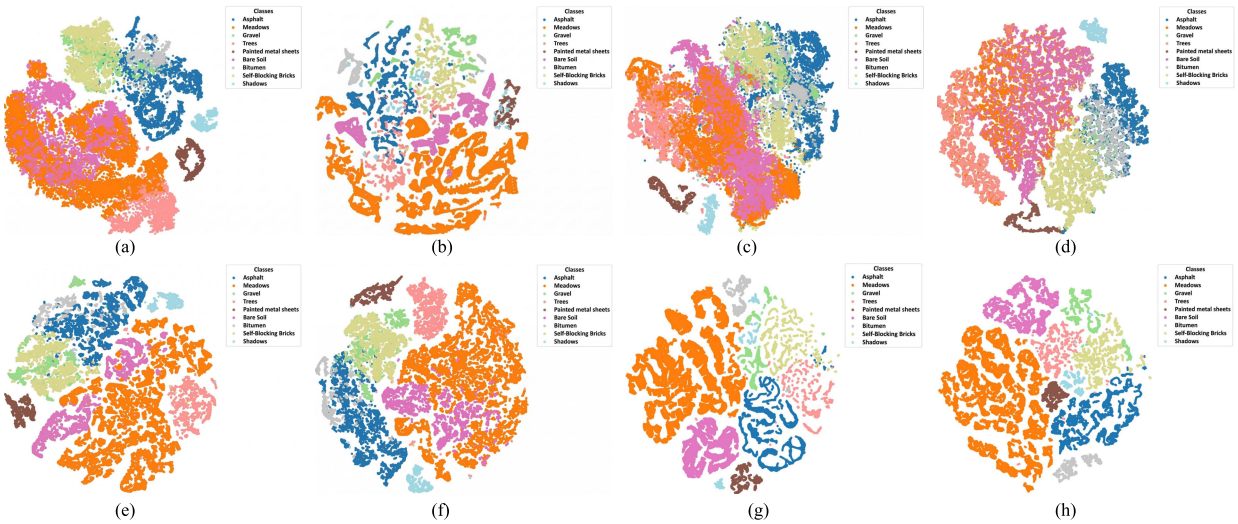
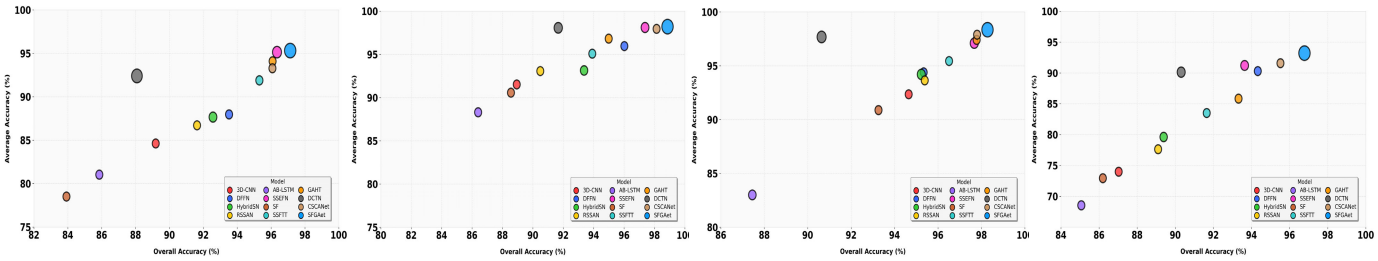


Fig. 16. T-SNE maps of different methods on the PU dataset. (a) 3D-CNN, (b) DFFN, (c) RSSAN, (d) AB-LSTM, (e) SSFTT, (f) GAHT, (g) CSCANet, (h) SFGANet.

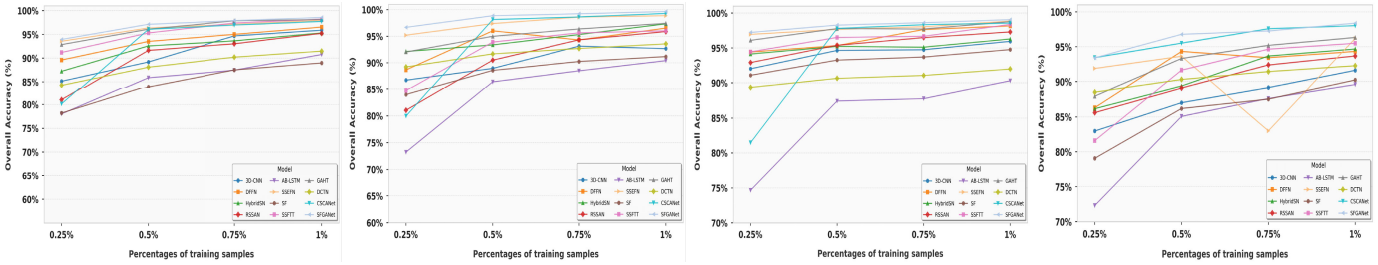
analysis using t-SNE visualization was conducted on the PU dataset. Fig. 16 shows a 2-D plot in which points of different colors represent samples from different classes. SFGANet yields highly compact clustering of sample points belonging to the same land cover class, with clear boundaries and no significant overlap between clusters of different land cover classes, which visually demonstrates the strong intraclass compactness and excellent interclass separability of the features extracted by the model and is well consistent with the quantitative classification accuracy metrics. For example, SFGANet achieved the tightest clustering results in the Meadows and Self-Blocking Bricks classes, which are the largest and most challenging classes. A possible reason for this is that DDFM, through its dual-path parallel architecture that combines global modeling

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(a) (b) (c) (d)

Fig. 17. Comparison of different models in terms of OA, AA, and training parameter count. (a) PU. (b) Salinas. (c) Xuzhou. (d) WHU-Hi-HanChuan.



(a) (b) (c) (d)

Fig. 18. Impact of different training sample percentages on OA. (a) PU. (b) Salinas. (c) Xuzhou. (d) WHU-Hi-HanChuan.

in the frequency domain with local refinement in the spatial domain, simultaneously captures the global spatial distribution patterns of objects and their local geometric texture details. On the other hand, AGIB achieves cross-modal deep interaction between the spatial and spectral branches, precisely preserving and enhancing local geometric structural information while reinforcing global semantic consistency. Ultimately, this yields excellent results characterized by tighter intraclass clustering and clearer interclass separation.

3) Efficiency Evaluation: To further assess the model efficiency, the total parameter count of different models was compared using detailed statistics as presented in Fig. 17. In the figure, the horizontal and vertical axes represent the OA and AA, respectively, while the radius of each circle denotes the order of magnitude of the model parameters. The results reveal that, although the proposed method has a relatively higher parameter count owing to the spatial–frequency domain feature interaction implemented by the DDFM, it achieves a favorable tradeoff between classification performance and model efficiency. In addition, to thoroughly evaluate the computational efficiency of the proposed method, we record the training and inference time consumption of all compared classification approaches, with the corresponding results summarized in Tables VI–IX. Overall, the proposed method not only exhibits superior classification accuracy but also delivers satisfactory performance in terms of practical computational efficiency.

4) Robustness Evaluation: To systematically evaluate the robustness of the proposed method, this study conducted comparative experiments on four datasets to assess classification accuracy under different sampling rates. For each dataset, 0.25%, 0.5%, 0.75%, and 1% of the samples were randomly selected for model training. The OA results for all methods are shown in Fig. 18. A comparative analysis of the results reveals that when training samples are extremely limited, some comparison methods exhibit a significant drop in accuracy, whereas the method proposed in this article consistently maintains high accuracy, fully demonstrating its exceptional robustness. Specifically, our method's classification performance was slightly inferior to that of the comparison methods in only one critical scenario: in the experiments on the WHU-Hi-HanChuan dataset with sampling rates of 0.25%–0.75%, its performance was slightly worse than that of CSCANet; in all other experimental scenarios, our method demonstrated the best classification performance. It is worth noting that under the 0.25% sampling rate, minority classes in large-scale datasets experienced extremely limited label conditions. For instance, in the WHU-Hi-HanChuan dataset, the model utilized only about six training samples for classes such as "Water-spinach" and "Bright-object." The ability of SFGANet to maintain leading performance in this scenario initially validates its robustness. This situation also occurs in both the Pavia University dataset and the Salinas dataset. However, if a strict extreme few-shot condition (e.g., five to ten samples per class) is imposed globally across all categories, the model encounters inevitable limitations. Under such absolute data scarcity, the data diversity becomes insufficient for the AGIB module to comprehensively capture complex cross-modal geometric alignments, which may lead to performance bottlenecks when distinguishing categories with highly overlapping spatial–spectral distributions.

E. Ablation Study

For verification of the effectiveness of the key components of the proposed SFGANet, ablation experiments targeted at five core modules (i.e., Conv3D, SBGM, DDFM, AGIB, and DDF) and the dual-branch architecture were conducted

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TABLE X

ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON THE PU DATASET. ("✓" INDICATES THAT THE COMPONENT IS RETAINED, AND "×" INDICATES THAT THE COMPONENT IS REMOVED)

Cases	Component						Indicators		
	Conv 3D	SBGM	DDFM	AGIB	DDF	Dual-Branch	OA(%)	AA(%)	Kappa(%)
1	×	✓	✓	✓	✓	✓	90.26 ± 01.14	86.41 ± 02.16	88.43 ± 01.16
2	✓	×	✓	✓	✓	✓	92.12 ± 01.25	85.14 ± 01.18	88.82 ± 01.41
3	✓	✓	×	✓	✓	✓	94.11 ± 01.12	91.73 ± 00.39	93.26 ± 01.24
4	✓	✓	✓	×	✓	✓	93.26 ± 01.21	89.12 ± 01.31	90.84 ± 00.68
5	✓	✓	✓	✓	×	✓	92.17 ± 02.41	86.49 ± 08.35	89.92 ± 05.22
6	✓	✓	×	×	✓	✓	90.16 ± 01.27	86.21 ± 02.25	87.63 ± 01.85
7	✓	✓	✓	✓	✓	×	89.23 ± 01.04	84.27 ± 02.37	86.61 ± 01.39
8	✓	✓	✓	✓	✓	✓	**97.13 ± 00.27**	**95.32 ± 00.52**	**95.91 ± 00.36**

TABLE XI

QUANTITATIVE COMPARISON OF DIFFERENT DECISION FUSION STRATEGIES (OA, %)

Cases	Datasets			
	PU	Salinas	Xuzhou	WHU-Hi-HanChuan
1	94.46 ± 01.18	93.85 ± 02.16	94.16 ± 00.39	93.17 ± 00.72
2	95.09 ± 00.76	95.71 ± 01.38	95.26 ± 00.35	93.94 ± 00.53
DDF	**97.13 ± 00.27**	**98.86 ± 00.36**	**98.28 ± 00.29**	**96.78 ± 00.28**

on the PU dataset. Three widely used metrics in HSI classification, including OA, AA, and K, were adopted to quantitatively evaluate the contributions of each component to the classification performance. The results are presented in Table X.

For clear analysis of the impact of each module on model performance and its suitability for hyperspectral data, the results of each case study are summarized in Table X. In Case 1, the removal of Conv3D led to a significant decrease in the OA of the PU dataset, indicating that Conv3D's simultaneous modeling of spectral and spatial correlation features is crucial for the preliminary extraction of hyperspectral features and acts as a core module for hyperspectral data processing. The removal of Conv3D weakened the model's ability to extract preliminary spatial–spectral features, which made it difficult to effectively capture the complex spatial and spectral information in hyperspectral data. In Case 2, the removal of the SBGM module resulted in a significant decline in AA, indicating that the SBGM module extracts spectral features from both the global statistical distribution and key significant responses through various pooling operations. The absence of SBGM caused the model to lose its ability to perform fine-tuned band weight calibration, which rendered it unable to flexibly adapt to the characteristic variations in hyperspectral data. In Case 3, after the DDFM was removed, the model lost its ability to extract global complementary priors in the frequency domain. At the same time, the model lacked a mechanism for selectively enhancing local spatial features, which resulted in a significant decline in the ability to jointly characterize spatial–spectral information in hyperspectral data and a corresponding decrease in classification accuracy. In Case 4, the classification accuracy of easily confused object categories—which exhibit highly similar spectral characteristics and complex spatial structures—declined the most after AGIB was removed, and this performance drop was further exacerbated in small-sample scenarios. Through the collaborative design of GSSA and a geometrically enhanced FFN, cross-branch semantic alignment and local geometric structure refinement are simultaneously achieved, with learnable parameters used to adaptively balance the two processes. The absence of AGIB prevented the effective integration of spatial and spectral features, which significantly weakened the model's ability to perform joint spatial–spectral representation. In Case 5, after the DDF was removed, significant performance degradation was observed. The DDF—as the core fusion unit at the model's output—can generate adaptive decision fusion weights for each pixel based on dual-branch semantic features. This design effectively mitigates overfitting in small-sample scenarios while preserving the independent discrimination capabilities of the spatial and spectral dual branches. In Case 6, DDFM and AGIB together form a complete pipeline that combines multidomain feature enhancement with cross-branch deep integration. The former establishes a high-quality feature foundation that balances global and local characteristics, while the latter achieves semantic alignment and bidirectional enhancement of spatial–spectral heterogeneous features. Together, they strengthen the model's ability to jointly represent spatial and spectral information, thereby mitigating overfitting issues in low-data-volume scenarios. In Case 7, removing the dual-branch architecture led to a significant drop in all core classification metrics, along with a marked increase in misclassification rates for confusable land cover classes and elevated small-sample overfitting risk. The original dual-branch design decouples spatial and spectral feature modeling, enabling specialized processing of these two heterogeneous feature types and providing the architectural foundation for AGIB's bidirectional cross-modal interaction and DDF. Without this architecture, the model fails to simultaneously extract precise spatial geometric structures and intrinsic spectral characteristics, weakening its joint spatial–spectral representation capability and resulting in performance degradation.

Furthermore, to quantitatively demonstrate the superiority of the proposed DDF mechanism over conventional late-fusion

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approaches, we conducted a direct comparison among three distinct decision-level fusion paradigms: in Case 1, constant average fusion, which uniformly assigns equivalent weights (i.e., a weight of 0.5 for each class) to the classification logits of both branches; in Case 2, global static weighting, which employs a pair of globally learnable scalar parameters to empirically optimize a fixed dataset-level fusion ratio; and the proposed DDF, which dynamically infers pixelwise adaptive fusion weights customized for the spatial and spectral representations of each specific sample. As shown in Table XI, while the static learned weights strategy slightly outperforms simple averaging by optimizing the global contribution ratio of the two branches, both approaches fall short of the proposed DDF. Because HSIs exhibit complex intraclass variability and interclass similarity, different pixels demand varying degrees of spatial and spectral information. By evaluating the decision confidence of the current pixel and generating instance-specific weights, DDF provides optimal semantic guidance for each sample individually, thereby achieving the best OA across all four datasets.

V. Conclusion

In this article, we propose SFGANet, a novel spectral–spatial–frequency hybrid CNN-Transformer for HSI classification. To extract robust representations, we introduce an SBGM to suppress intraclass spectral variance. This is complemented by a DDFM, which synergizes global frequency-domain long-range modeling via 2D-RFFT with local spatial texture refinement. Furthermore, an AGIB establishes deep cross-modal semantic alignment between spatial and spectral features. Finally, to combat overfitting in few-shot scenarios, a DDF strategy adaptively aggregates dual-branch predictions using pixel-level generated weights. Experiments on four public datasets demonstrate the effectiveness and generalization capability of this method. Despite its outstanding classification performance, the proposed architecture has an inherent limitation when the available training labels are extremely restricted (e.g., five to ten samples per class), as the deep spatial–frequency semantic modeling relies on a baseline threshold of data diversity. To overcome this diversity bottleneck, our future work will focus on developing lightweight optimization schemes that reduce the model's parameter dependency. Furthermore, to explicitly address extreme label scarcity, we plan to extend this architecture by integrating semisupervised, unsupervised, and metalearning paradigms, thereby leveraging abundant unlabeled spatial–spectral context to compensate for the lack of annotated diversity.

References

[1] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[2] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[3] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[4] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[5] W. Xie, J. Zhang, J. Lei, Y. Li, and X. Jia, "Self-spectral learning with GAN based spectral–spatial target detection for hyperspectral image," *Neural Netw.*, vol. 142, pp. 375–387, Oct. 2021.

[6] L. Wang and P. Wang, "Sub-pixel mapping for hyperspectral imagery using super-resolution then spectral unmixing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 461–464.

[7] F. Kizel, M. Shoshany, N. S. Netanyahu, G. Even-Tzur, and J. A. Benediktsson, "A stepwise analytical projected gradient descent search for hyperspectral unmixing and its code vectorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 4925–4943, Sep. 2017.

[8] Z. Lv, T. Yang, P. Zhong, W. Sun, J. A. Benediktsson, and J. Li, "Hierarchical feature fusion triple network for change detection with bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4401512.

[9] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[10] C. Shi, K. Shi, F. Zhu, Z. Zeng, and L. Wang, "A multilevel domain similarity enhancement guided network for remote sensing image compression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5645819.

[11] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[12] C. Fu, B. Du, and L. Zhang, "ReSC-Net: Hyperspectral image classification based on attention-enhanced residual module and spatial-channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518615.

[13] L. Wang, H. Wang, S. Yin, and L. Wang, "Masked vision transformer for fast hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5212216.

[14] J. Liao and L. Wang, "HierarchicalMamba: A multiscale state space model with dual transitions for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5402025.

[15] N. Victor et al., "Remote sensing for agriculture in the era of Industry 5.0—A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5920–5945, 2024.

[16] J. E. Patino and J. C. Duque, "A review of regional science applications of satellite remote sensing in urban settings," *Comput., Environ. Urban Syst.*, vol. 37, pp. 1–17, Jan. 2013.

[17] Y. Yang, K.-S. Chen, J. Atli Benediktsson, and M. Örn Úlfarsson, "Physical interpretation of microwave emission from snow-covered stratified sea ice with rough boundaries," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 2001217.

[18] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022.

[19] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[20] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[21] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2012.

[22] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[23] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

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[24] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[25] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[26] T. Wen, H. Wang, and L. Wang, "Dual-branch spatial–spectral transformer with similarity propagation for hyperspectral image classification," *Remote Sens.*, vol. 17, no. 14, p. 2386, Jul. 2025.

[27] Y. Cui, M. Jia, L. Wang, S. Gao, L. Chen, and C. Zhao, "Hyperspectral image classification based on normalized grouping fused comprehensive relative position transformer network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 1515–1532, 2026.

[28] J. Zhou et al., "Exploring multi-timestep multi-stage diffusion features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519516.

[29] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient Mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.

[30] J. Qu, J. Zhao, W. Dong, L. Zhang, and Y. Li, "MambaFedCD: Spatial–spectral–temporal collaborative mamba-based active federated hyperspectral change detection," *IEEE Trans. Image Process.*, vol. 35, pp. 1478–1492, 2026.

[31] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.

[32] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[33] Z. Lv, P. Zhang, P. Wang, W. Sun, M. Zhao, and R. Zhu, "Novel sample augmentation approach for improving classification performance with high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5643713.

[34] N. Chen, J. Yue, L. Fang, and S. Xia, "SpectralDiff: A generative framework for hyperspectral image classification with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522416.

[35] T. Zhang, C. Xuan, F. Cheng, Z. Tang, X. Gao, and Y. Song, "CenterMamba: Enhancing semantic representation with center-scan mamba network for hyperspectral image classification," *Expert Syst. Appl.*, vol. 287, Aug. 2025, Art. no. 127985.

[36] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[37] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[38] M. Ahmad, U. Ghous, M. Usama, and M. Mazzara, "WaveFormer: Spectral–spatial wavelet transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[39] G. Miao, Z. Xie, and F. Yang, "Wavelet-enhanced spatial–spectral prior injection mamba network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5501117.

[40] B. Shi, G. Cao, Y. Zhang, Y. Liu, and K. Yang, "Fourier-based spectral–spatial generator for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5514417.

[41] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[42] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[43] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[44] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[45] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[46] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[47] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[48] Y. Qing, W. Liu, L. Feng, and W. Gao, "Improved transformer net for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 11, p. 2216, Jun. 2021.

[49] X. He, Y. Chen, and Q. Li, "Two-branch pure transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[50] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513.

[51] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[52] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

[53] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[54] W. Qi, C. Huang, Y. Wang, X. Zhang, W. Sun, and L. Zhang, "Global–local 3-D convolutional transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510820.

[55] L. Sun, H. Zhang, Y. Zheng, Z. Wu, Z. Ye, and H. Zhao, "MASSFormer: Memory-augmented spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516415.

[56] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "DCTN: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.

[57] B. Dun, D. Liu, L. Wang, and J. A. Benediktsson, "Dual-branch token selection fusion network for hyperspectral image classification with few labeled samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 8965–8983, 2026.

[58] Y. Ding et al., "Multi-scale receptive fields: Graph attention neural network for hyperspectral image classification," *Expert Syst. Appl.*, vol. 223, Aug. 2023, Art. no. 119858.

[59] Z. Zhang et al., "Multireceptive field: An adaptive path aggregation graph neural framework for hyperspectral image classification," *Expert Syst. Appl.*, vol. 217, May 2023, Art. no. 119508.

[60] Y. Ding et al., "SLCGC: A lightweight self-supervised low-pass contrastive graph clustering network for hyperspectral images," *IEEE Trans. Multimedia*, vol. 27, pp. 8251–8262, 2025.

[61] Y. Ding et al., "AF2GNN: Graph convolution with adaptive filters and aggregator fusion for hyperspectral image classification," *Inf. Sci.*, vol. 602, pp. 201–219, Jul. 2022.

[62] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[63] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[64] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[65] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[66] S. Liu, C. Fu, Y. Duan, X. Wang, and F. Luo, "Spatial–spectral enhancement and fusion network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5502414.

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[67] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.
[68] B. Zhang, Y. Chen, S. Xiong, and X. Lu, "Hyperspectral image classification via cascaded spatial cross-attention network," *IEEE Trans. Image Process.*, vol. 34, pp. 899–913, 2025.
[69] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8024–8035.
[70] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.



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